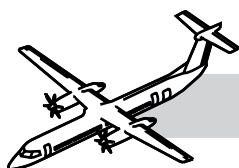


CHAPTER 45

CENTRAL DIAGNOSTIC SYSTEM

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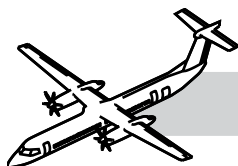
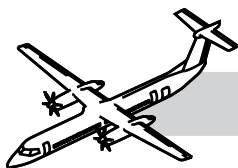


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CHAPTER 45

CENTRAL DIAGNOSTIC SYSTEM

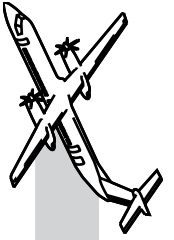


45-45-00 CENTRAL DIAGNOSTIC SYSTEM

INTRODUCTION

The Central Diagnostic System (CDS) is an On-Board Maintenance System. It supports maintenance personnel by monitoring, controlling, recording and displaying system tests, diagnostics and maintenance activities.

The systems which interface with the CDS may communicate to CDS via digital buses or discrete signals and they may operate with different levels of built-in-test capabilities. As a result, there are differences in the individual system menus and their contents.



45-2

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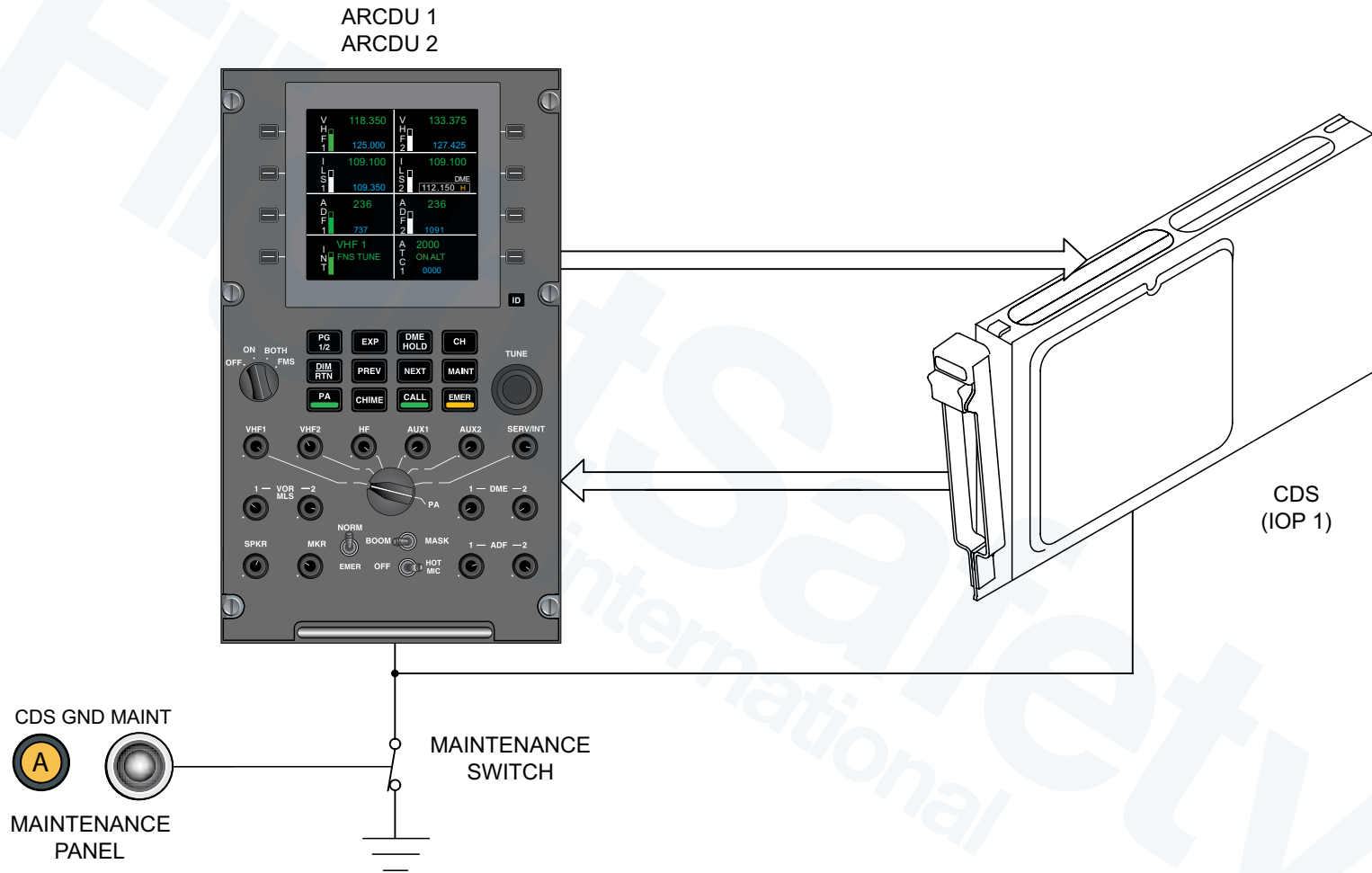
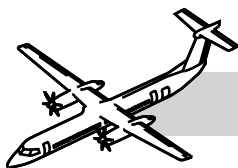


Figure 45-1. Central Diagnostic System Block Diagram



GENERAL

Refer to:

- Figure 45-1. Central Diagnostic System Block Diagram.
- Figure 45-2. CDS Architecture.

The CDS uses the input/output processor module 1 (IOP 1) and input/output module 1 (IOM 1) in the integrated flight cabinet 1 (IFC1), and the two audio radio control and display units (ARCDU1 and ARCDU2).

The CDS is physically located in IOP module 1 and uses the same processor as the flight data processing system (FDPS). There are two partitions in the IOP 1 module, one for the CDS and one for the FDPS.

The IOP has an ARINC extension card to handle all the ARINC inputs.

Discrete inputs are received through the IOM 1 and the ARINC 429 inputs are received through the ARINC extension card, or directly to the CPU card of the IOP 1.

Maintenance data is supplied to the CDS from 3 different types of systems:

Type 1 systems: The type 1 systems interface with the CDS with one ARINC 429 input and one ARINC 429 output buses to supply commands and responses. The type 1 systems continue to operate in their operational mode until the CDS supplies a request to the system to start the maintenance mode. Type 1 systems supply the following data to the CDS as follows:

- System status (health) monitor
- LRU identification (hardware and software part numbers)
- Initiated built-in test (IBIT).

Type 2 systems: The type 2 systems interface with the CDS with one ARINC 429 output bus and a discrete input to start the IBIT.

Type 2 systems supply the following data to the CDS:

- System status (health) monitor
- LRU identification (hardware and software part numbers)
- Initiated built-in test (IBIT).

Type 3 systems: Type 3 systems only supply their health data to the CDS. The type 3 systems cannot do an IBIT.

NOTE

The IBIT request affects the safety of the aircraft and cannot be done in flight.

The avionic systems also supply the following data to the CDS:

- Internal fault
- Interface fault
- Probe fault (external, does not interface directly with the CDS)
- Class 2 fault
- Class 1 fault
- System in test (Power-on self test or IBIT)
- Command response (maintenance mode).

Class 1 failures are failures that prevent continued operation of an LRU or channel and are annunciated via advisory messages, cautions, warnings or advisory lights in the flight compartment. Dispatch with such posted failures is to be in accordance with the Master Minimum Equipment List (MMEL). Class 2 failures are failures which do not prevent continued system operation and will not be annunciated to the flight crews. Such faults would be evident during maintenance interrogation, which is performed on opportunity basis. Class 2 faults do not affect dispatch and will be listed in the Fault Isolation Manual (FIM).

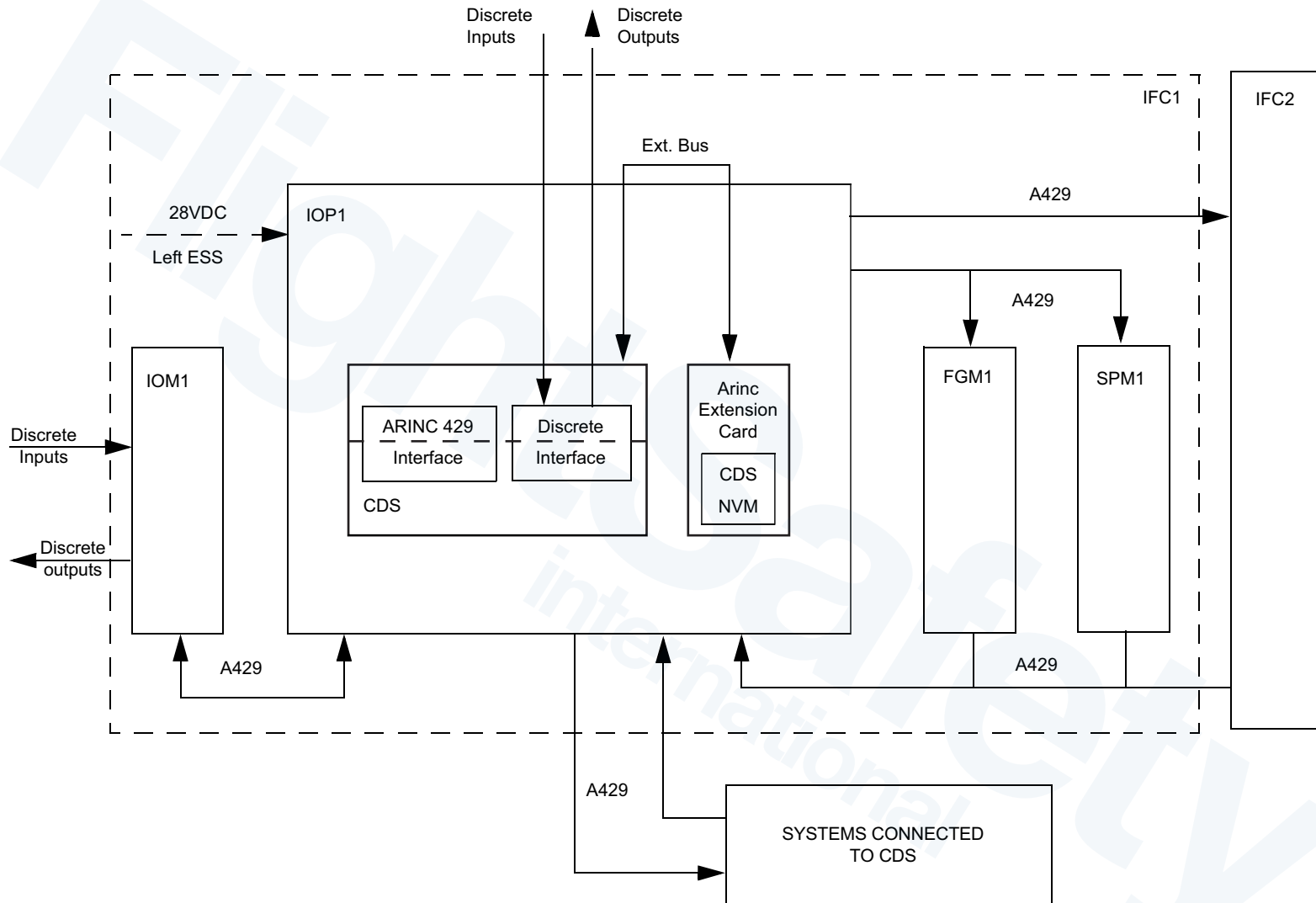
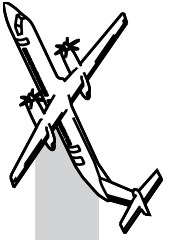
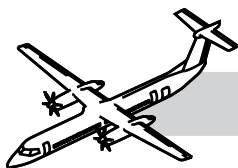


Figure 45-2. CDS Architecture



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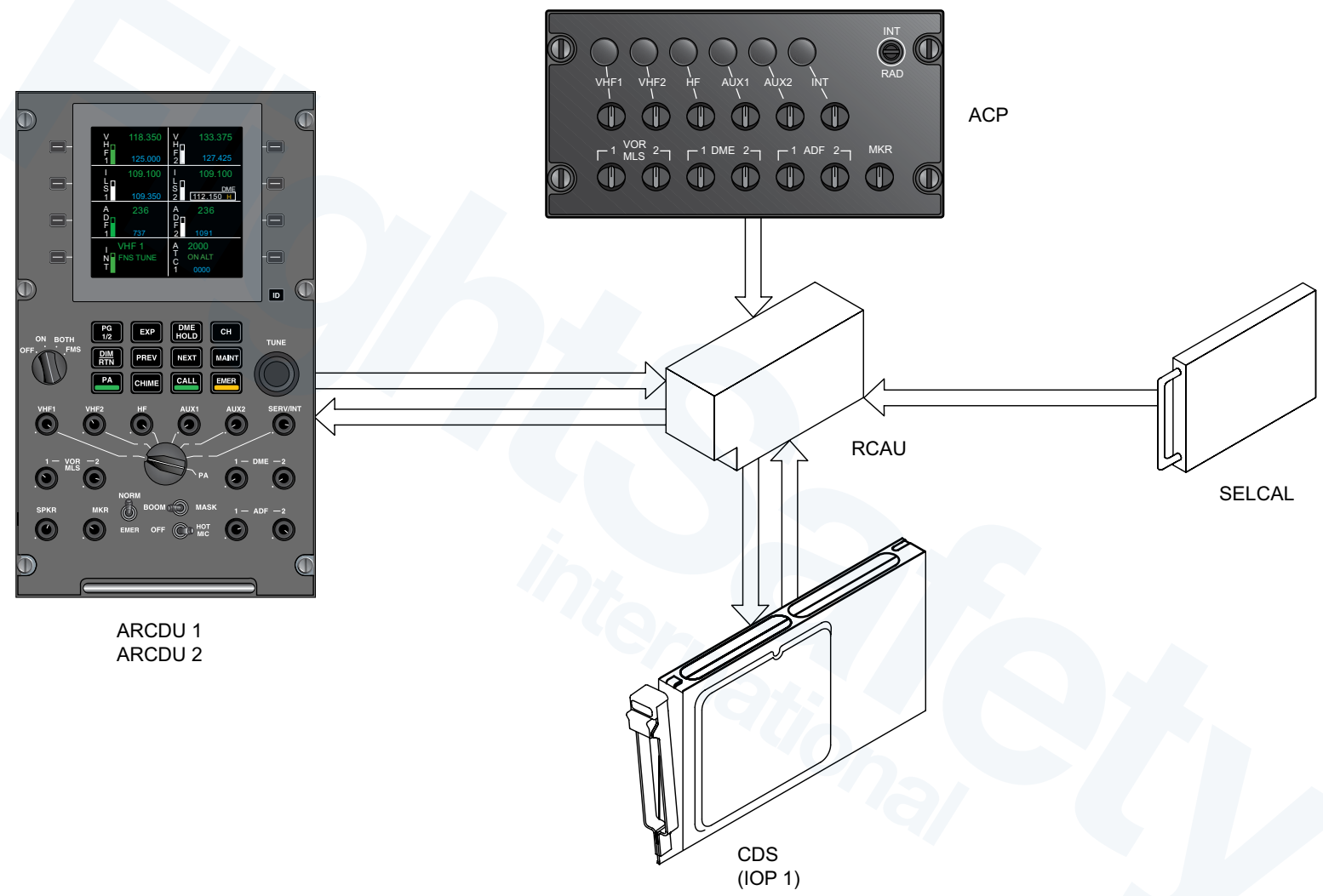
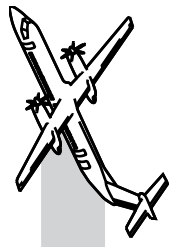
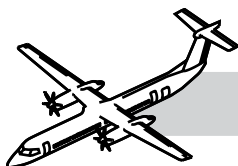


Figure 45-3. Interface with ARMS (Type 1 Avionics System)



Refer to Figure 45-3. Interface with ARMS (Type 1 Avionics System).

NOTES

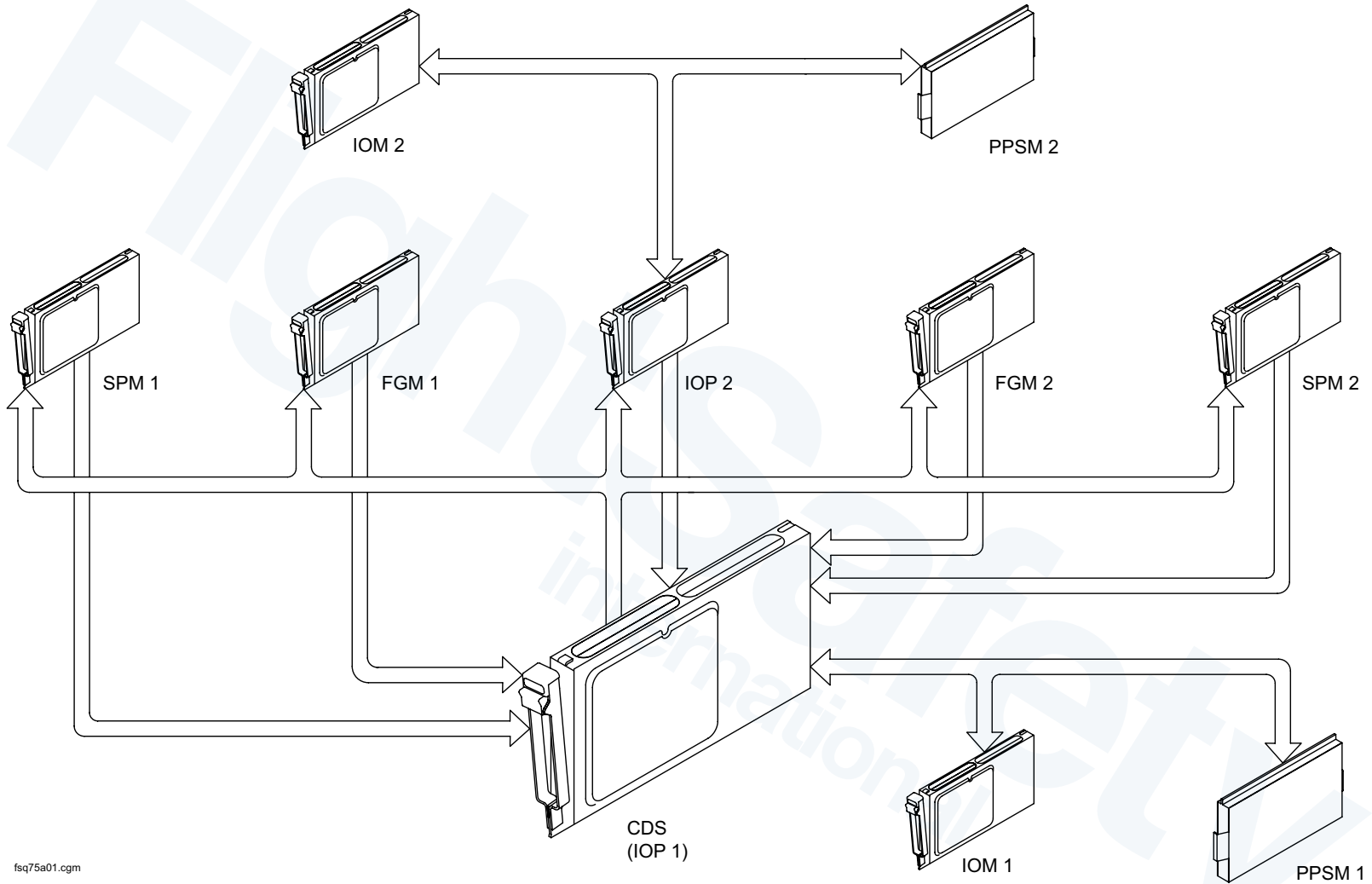
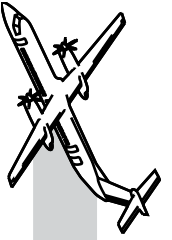
The CDS is part of the Thales Avionics Suite, which is comprised of autoflight, navigation, communication and electronic display equipment. This group of equipment records system faults as they occur as they are directly monitored by the IOP, thereby allowing the CDS to see equipment status in real-time. This also allows the failures to be time-stamped. Due to the real-time monitoring, manual clearing of stored faults is not necessary for equipment within the “AVIONICS” menu.

The Audio Radio Management System (ARMS) has the units identified as follows:

- RCAU
- ARCDU1, probe sensed by the pilot's audio card in the RCAU
- ARCDU2, probe sensed by the co-pilot's audio card in the RCAU
- Observer's ACP, probe sensed by the observer's audio card in the RCAU
- Optional SELCAL, probe sensed by the RCAU.

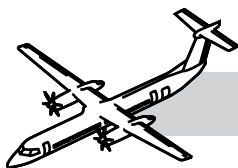
NOTE

A probe is any LRU contained in the system that cannot report directly to CDS.



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Figure 45-4. Interface with Type 1 Avionics Systems Installed within the IFCs



Refer to Figure 45-4. Interface with Type 1 Avionics Systems Installed within the IFCs.

NOTES

IFC1 contains:

- FGM 1
- Stall Protection Module 1 (SPM 1)
- IOP 1
- PPSM 1
- IOM 1.

IFC2 contains:

- FGM 2
- SPM 2
- IOP 2
- PPSM 2
- IOM 2.

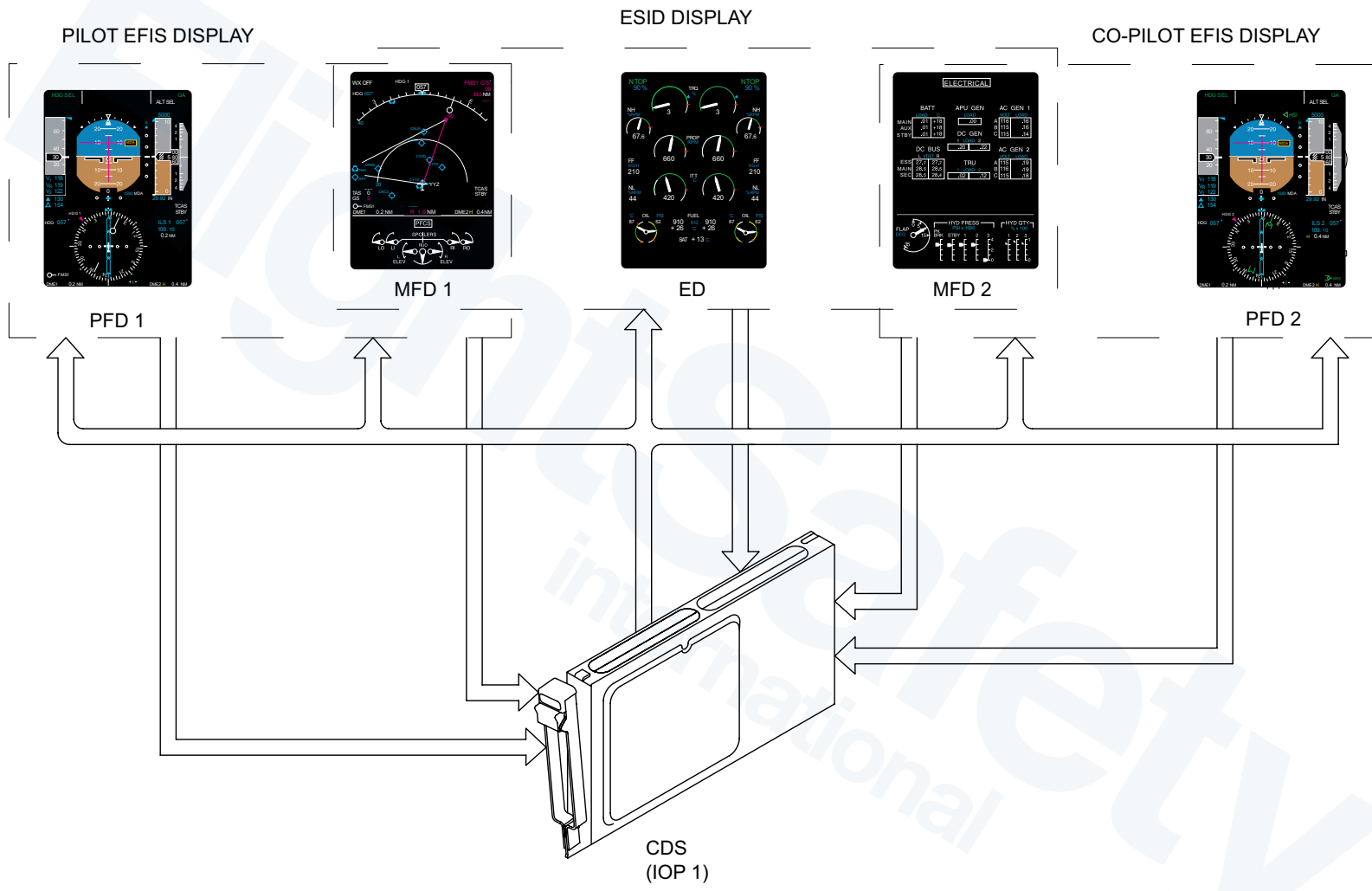
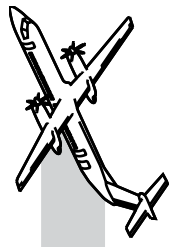
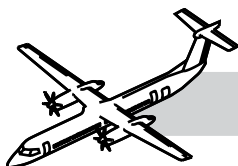


Figure 45-5. Interface with the Electronic Instrument System EIS (Type 1 Avionics Systems)

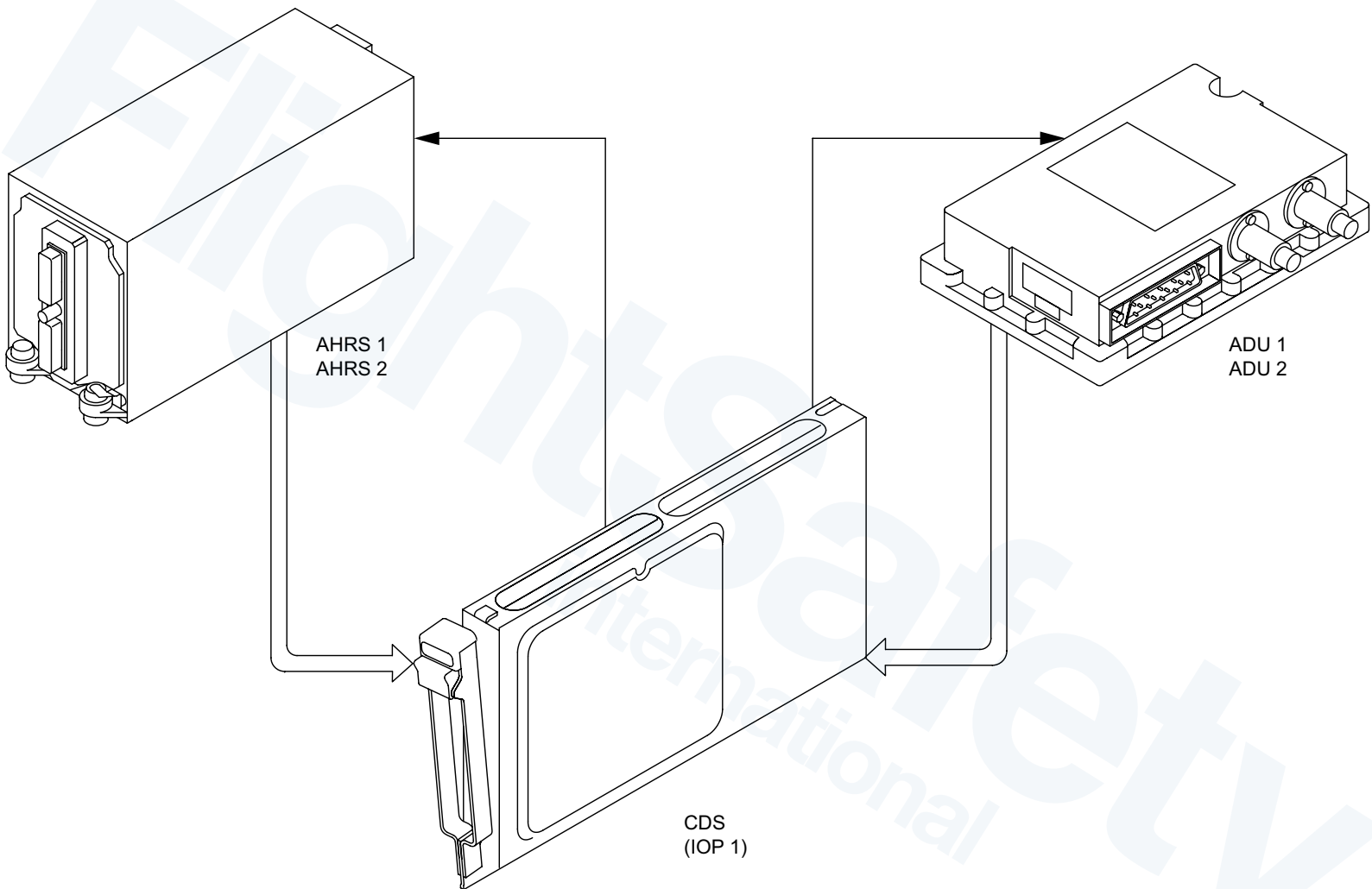
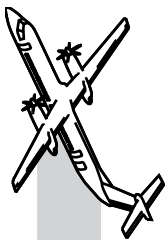


Refer to Figure 45-5. Interface with the Electronic Instrument System EIS (Type 1 Avionics Systems).

NOTES

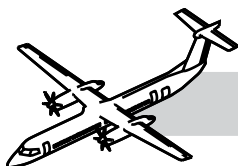
The EIS has 5 type 1 display units identified as:

- PFD 1
- PFD 2
- MFD 1
- MFD 2
- ED.



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Figure 45-6. Interface with Avionics Type 2 Systems



Refer to Figure 45-6. Interface with Avionics Type 2 Systems.

NOTES

The avionics type 2 units are:

- ADU 1
- ADU 2
- AHRS 1
- AHRS 2.

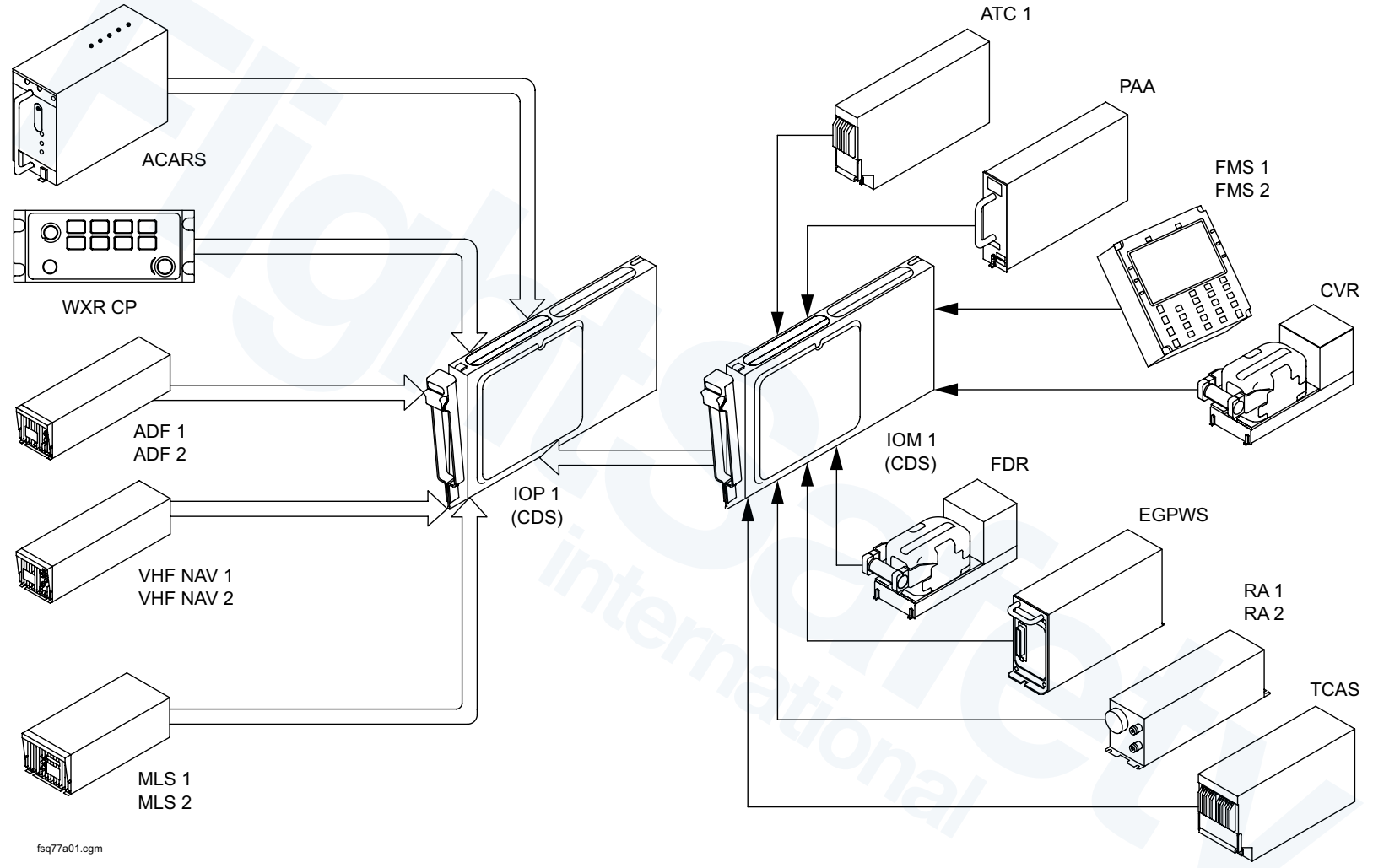
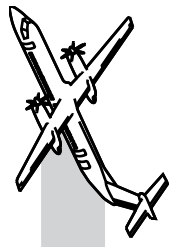
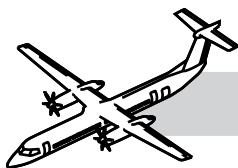


Figure 45-7. Interface with Avionics Type 3 Systems



Refer to Figure 45-7. Interface with Avionics Type 3 Systems.

NOTES

The avionics type 3 systems are identified as:

- ACARS
- CVR
- FDR
- RA
- EGPWS
- ATC
- ADF
- VHF NAV
- TCAS
- PAA
- FMS and GPS
- Weather Radar Control Panel.

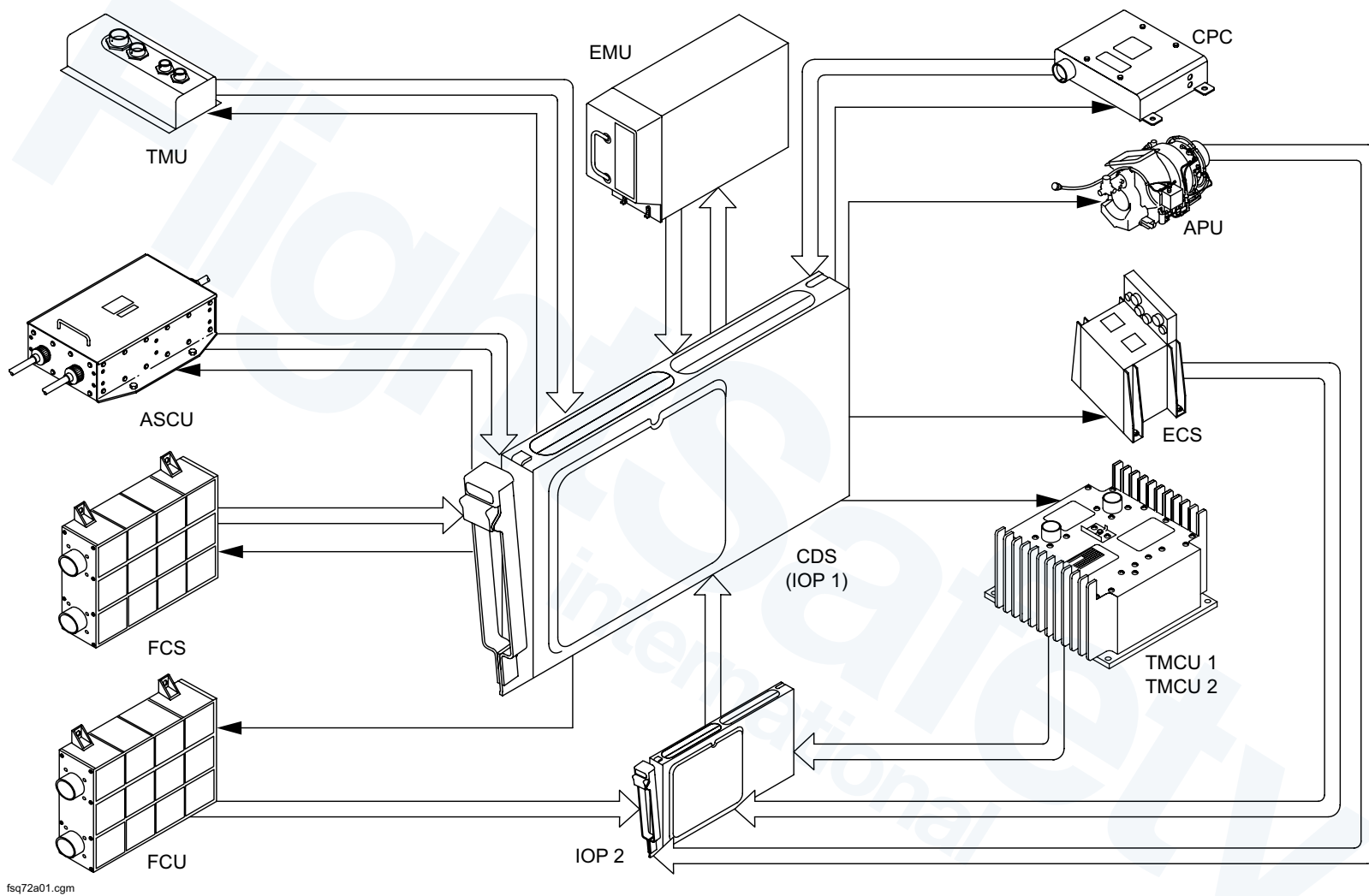
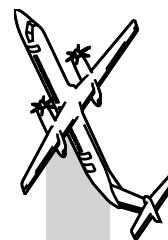
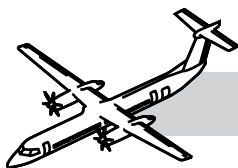


Figure 45-8. Interface with Other (Non Avionics) Systems



Refer to Figure 45-8. Interface with Other (Non Avionics) Systems.

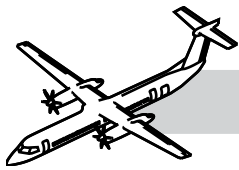
NOTES

There is a selected group of electronic control units, referred to as “OTHER SYSTEMS”. These ECUs from the various aircraft systems internally store their own respective system faults in Non-Volatile Memory (NVM), retaining such information through all power interruptions for recall via CDS interrogation by maintenance personnel.

Most of the units within this category use the “transmit data request” discrete to initiate the one-shot “data dump” to the CDS. The “transmit data request” is output from the CDS by the action of menu key selection to the desired system. Once dumped into the CDS temporary RAM, the ECU remains idle and the CDS interacts solely with the ARCDU. If an update of the displayed menu contents is desired, such as after an adjustment or fault rectification, the system menu must be exited and then re-entered to initiate a new “transmit data request”.

The other aircraft system units that are connected to the CDS are identified as:

- Engine Monitoring Unit (EMU)
- Timer Monitor Control Unit 1 (TMCU 1)
- Timer Monitor Control Unit 2 (TMCU 2)
- Auxiliary Power Unit Full Authority Electronic Engine Control (APU FADEC)
- Flap Control Unit (FCU)
- Flight Control Electronic Control Unit (FCECU)
- Environmental Control System Electronic Control Unit (ECSECU)
- Cabin Pressure Controller (CPC)
- Anti Skid Control Unit (ASCU).

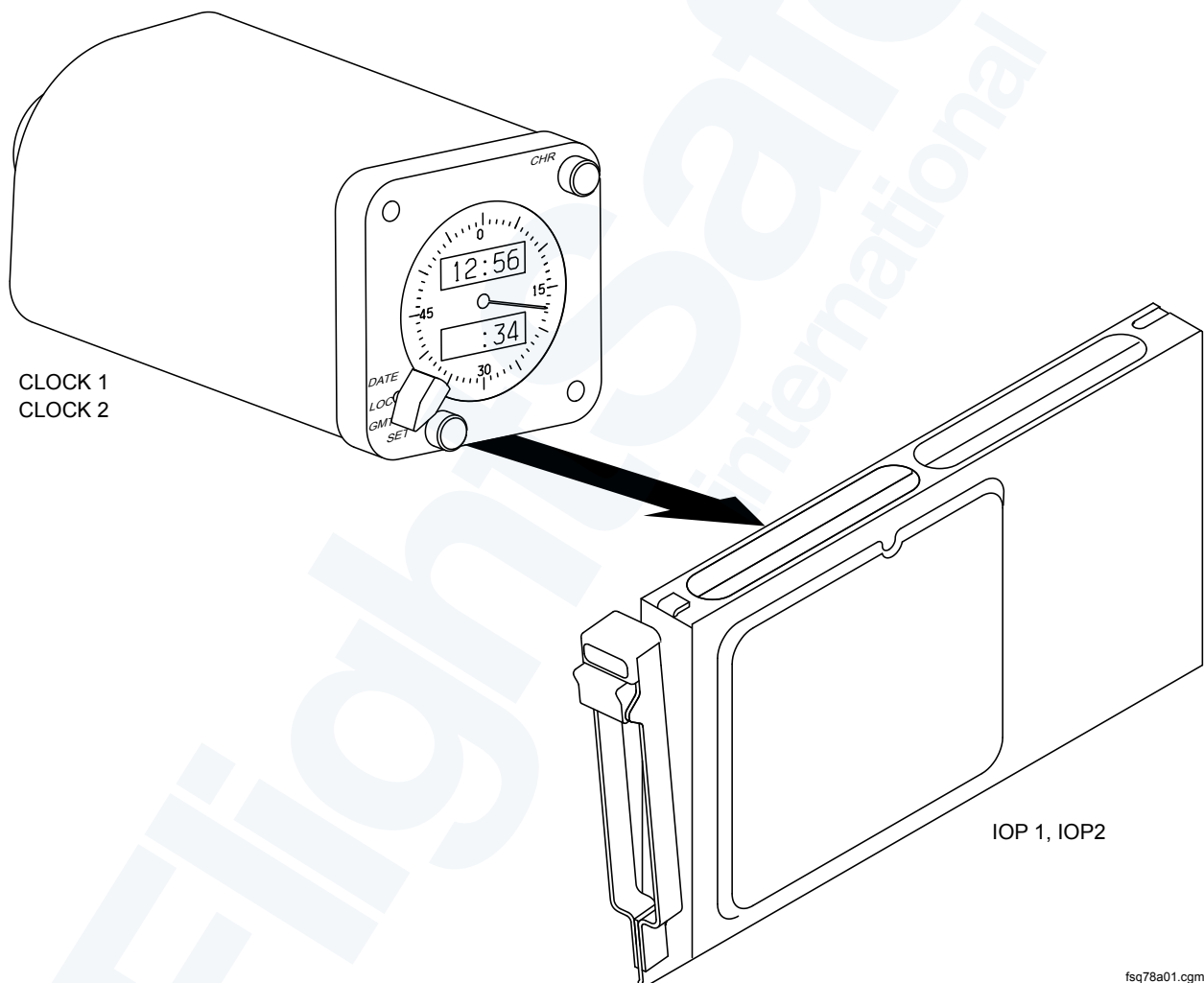


Refer to Figure 45-9. Interface with Clocks.

The two clocks supply date and time through the FDPS to the CDS.

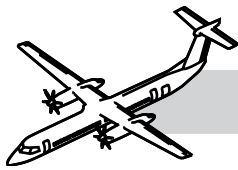
NOTE

The clocks are not monitored for faults by the CDS.

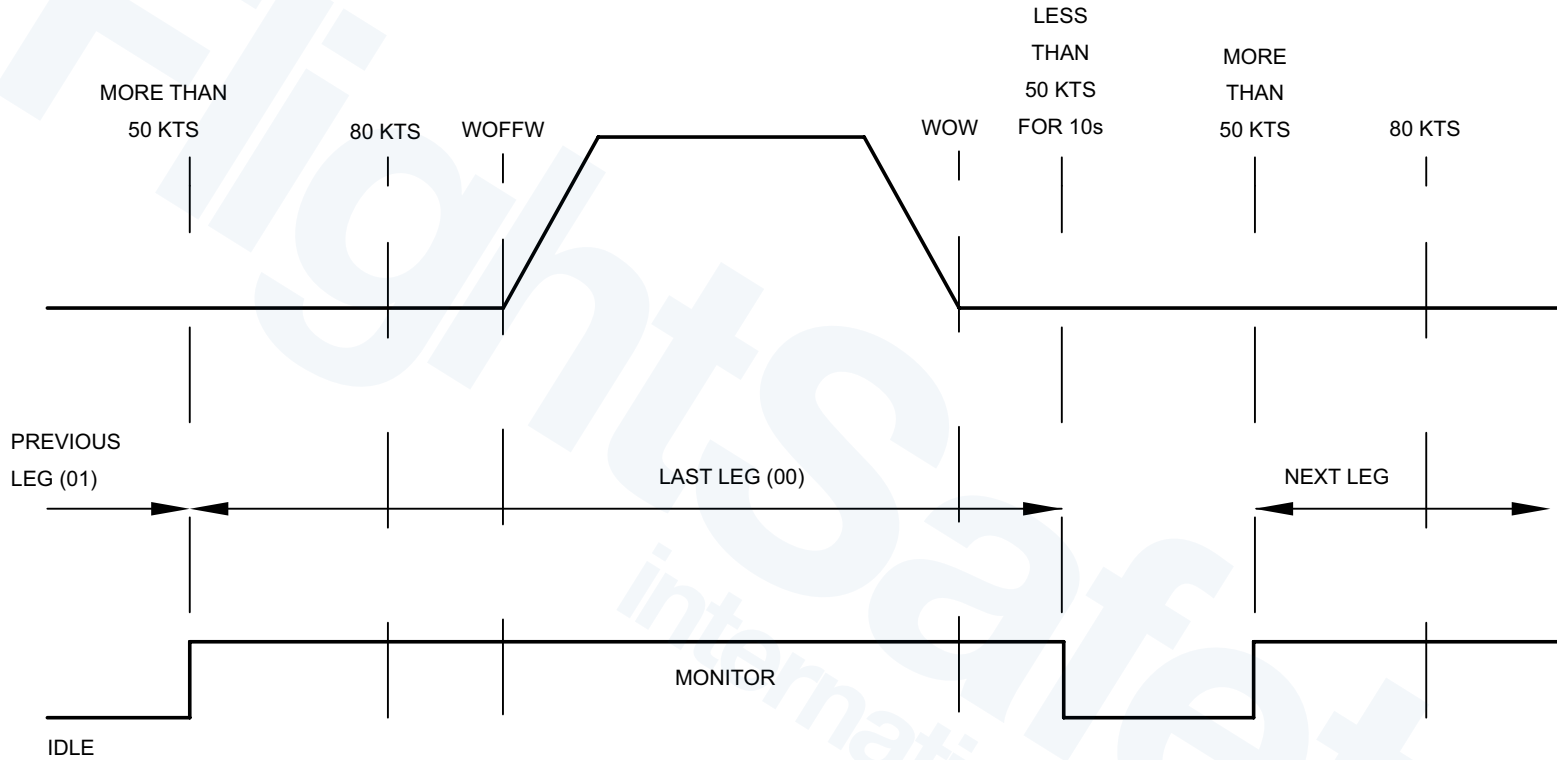
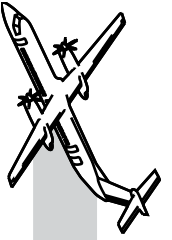


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Figure 45-9. Interface with Clocks

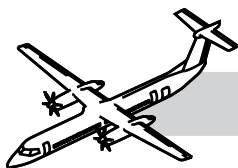


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Figure 45-10. Modes of Operation



SYSTEM DESCRIPTION

Refer to Figure 45-10. Modes of Operation.

The CDS operates in the following modes:

Normal Monitoring Mode: During the normal monitoring mode, the avionic systems continuously supply their fault condition to the CDS.

The normal monitoring mode is set when one of these conditions is sensed:

- Aircraft is weight off wheels or the Calculated Air Speed is more than 50 knots
- Aircraft is weight on wheels and the CDS maintenance switch is not set on.

In the normal mode the CDS has the functions that follow:

- Transmits fault data (date, time and weight on wheels conditions to the type 1 systems)
- Transmits aircraft type and serial number to the remote control audio unit (RCAU)
- Receives system status data from other avionic systems
- Receives the avionic configuration data
- Stores fault reports
- Sorts status data from the different systems
- Supplies fuel flow, fuel temperature, and main oil pressure parameters to the engine monitoring unit (EMU).

When the CAS is less than 50 knots, the system monitors for internal failures only and, if a malfunction is sensed, it is saved in the ground memory of the unit and is transmitted to the CDS.

When the CAS is between 50 and 80 knots, the system monitors for internal failures only and, if a malfunction is sensed, it is saved in the flight memory of the unit and is transmitted to the CDS.

When the CAS is more than 80 knots, the system monitors for internal and external failures. If a malfunction is sensed, it is saved in the flight memory of the unit, and is transmitted to the CDS.

Class 2 malfunctions sensed during the last leg are erased at the start of the new one.

Maintenance mode: One of the two ARCDUs is used to do initiated built-in tests (IBITs) and see the avionic system status, LRU identification, and configuration.

The maintenance mode is set when all of the conditions are as follows:

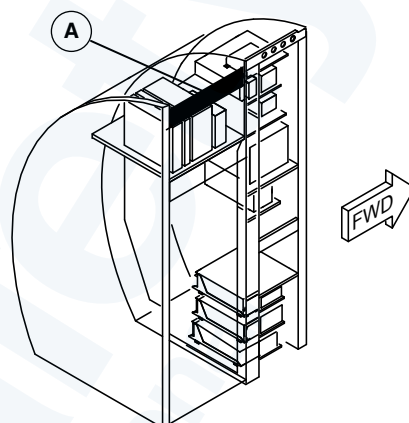
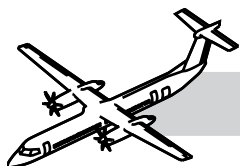
- Aircraft is weight on wheels and CAS is less than 50 knots for more than 10 sec
- CDS maintenance switch on the maintenance panel is set
- Maintenance key on the ARCDU is set.

Test: The test of the CDS and IOP 1 is done by the flight data processing system (FDPS).

Idle mode: During the idle mode, the avionic systems do not supply their fault condition to the CDS.

The idle mode is set when all the conditions are as follows:

- Aircraft is on the ground
- CDS maintenance switch on the maintenance panel is set.



**AVIONICS RACK ELECTRICAL
INSTALLATION**

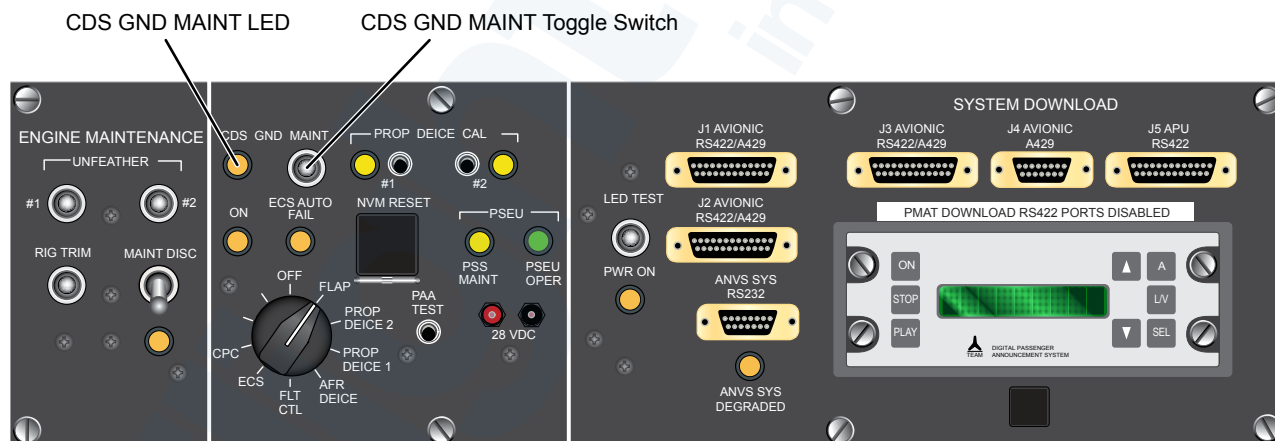
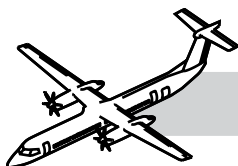


Figure 45-11. Maintenance Panel, CDS Maintenance Mode Selection



COMPONENT DESCRIPTION

CDS Maintenance Switch

Refer to Figure 45-11. Maintenance Panel, CDS Maintenance Mode Selection.

The CDS maintenance switch, installed on the maintenance panel located above the wardrobe compartment, sets the maintenance mode for the CDS.

CONTROLS AND INDICATIONS

Refer to Figure 45-11. Maintenance Panel, CDS Maintenance Mode Selection.

The maintenance mode is set when:

- Aircraft is weight on wheels and CAS is less than 50 knots for more than 10 seconds and
- CDS GRD MAINT switch is set.

The amber CDS GRD MAINT light on the maintenance panel comes on.

Operational Check After The CDS Maintenance Switch Has Been Replaced

Refer to the Bombardier AMM PSM 1-84-2 for a detailed description of this maintenance practice.

- The check is to ensure that the ARCDU can be switched over to display the maintenance page after the maintenance switch is replaced.

Return to Service Operational Check after the Maintenance Panel Has Been Replaced

Refer to the Bombardier AMM PSM 1-84-2 for a detailed description of this maintenance practice.

The return to service operational check is to ensure that all the features and indication on the panel continue to work:

- CDS GND MAINT light comes on when MAINT toggle is selected to the UP position
- PAA TEST switch is pushed to ensure the hi-lo chime is heard through the cabin speakers
- PWR ON light of the SYSTEM DOWNLOAD section comes ON when the PWR ON toggle switch is set to "PWR ON" position. It should turn off when PWR ON is deselected
- MAINT DISC light comes on when MAINT DISC toggle is selected in the "up" position
- MAINT key on left ARCDU is pushed to confirm that the CDS menu appears, repeat for the right ARCDU
- Push the DIM/RTN key and check the CDS main menu is displayed
- Push the CDS GND MAINT toggle to the down position and verify the CDS GND MAINT light is OFF and pushing the MAINT key on the ARCDU would not bring up the CDS main menu page.

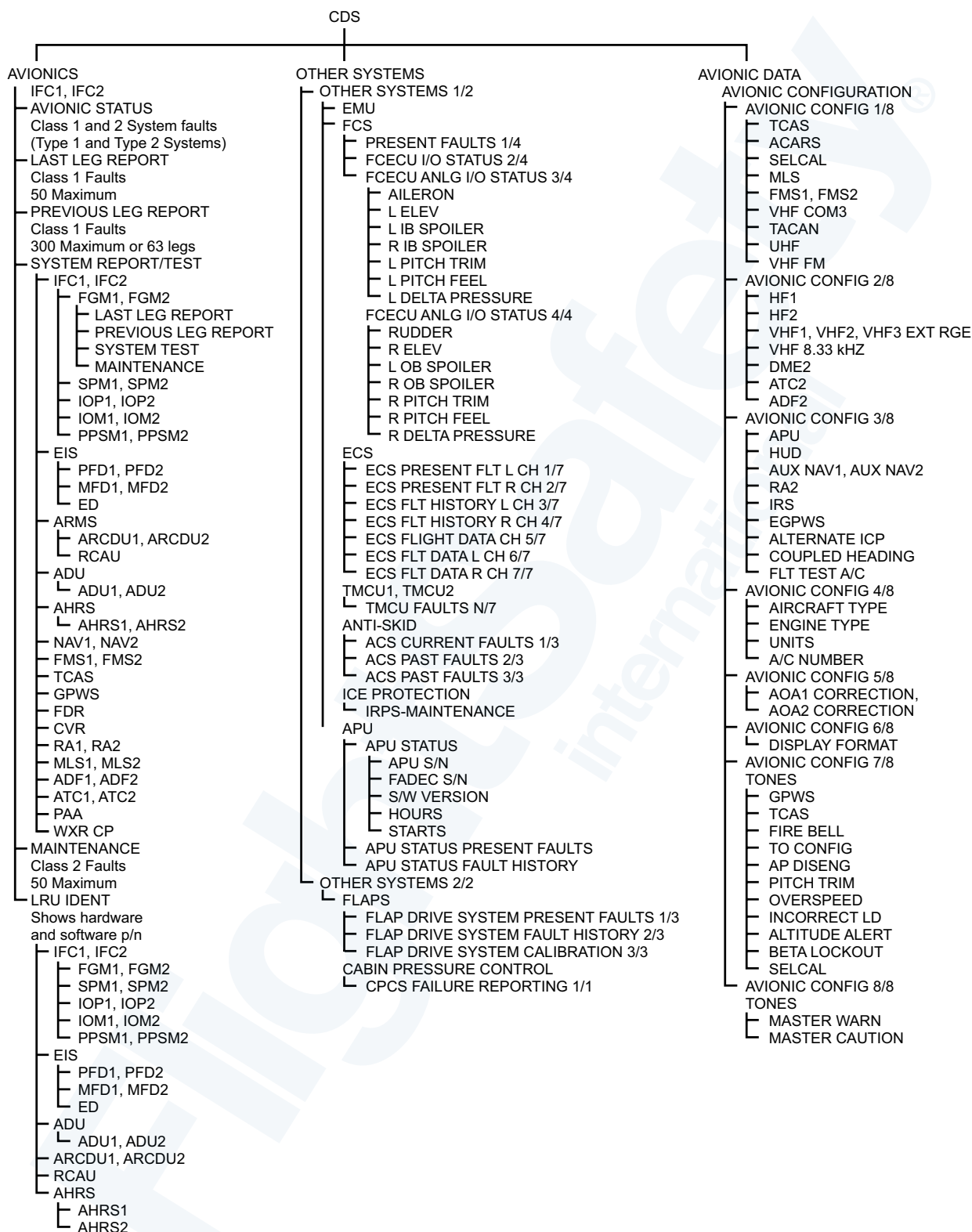
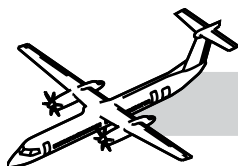
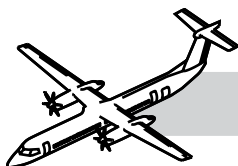


Figure 45-12. CDS Pages



Refer to Figure 45-12. CDS Pages.

NOTES

See Figure 45-12 for a list of the CDS pages available within the three separate categories:

- AVIONICS
- OTHER SYSTEMS
- AVIONIC DATA.

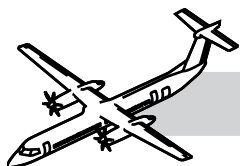
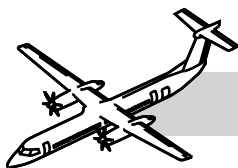


Figure 45-13. ARCCDU, CDS Maintenance Mode Selection



Refer to Figure 45-13. ARCDU, CDS Maintenance Mode Selection.

The MAINT key on an ARCDU is pushed to continue the sequence to set the CDS maintenance mode.

The CDS maintenance mode is stopped with any of the following conditions:

- System is de-energized
- RTN key on the ARCDU is pushed
- Logoff (communication malfunction between the ARCDU and CDS)
- One of the conditions used to set the CDS maintenance mode is not correct.

Only one ARCDU can show the pages for maintenance mode at a time. When one ARCDU is set, the other has no effect until the communication with the other ARCDU is stopped.

The ARCDU has the controls for the CDS that follows:

- 8 side keys
- Return (RTN)
- Previous (PREV)
- NEXT
- Maintenance (MAINT).

NOTE

Each key has only one function except for the engine monitoring unit (EMU). During some operations the PREV and RTN key perform the same function.

A command is supplied from the ARCDU to the CDS to start the maintenance mode.

When the CDS receives the command, it transmits a request to send (RTS) signal back to the ARCDU. When the ARCDU is ready to receive the data, a clear to send (CTS) signal is supplied back to the CDS.

The CDS main menu page is shown if the CDS is operating correctly.

A CDS NOT AVAILABLE message is shown by the ARCDU if no response is received from the CDS after 3 seconds.

A SYSTEM NOT AVAILABLE message is shown if no response is received by the CDS after a selection of a system and the CDS does not send any data to the ARCDU.

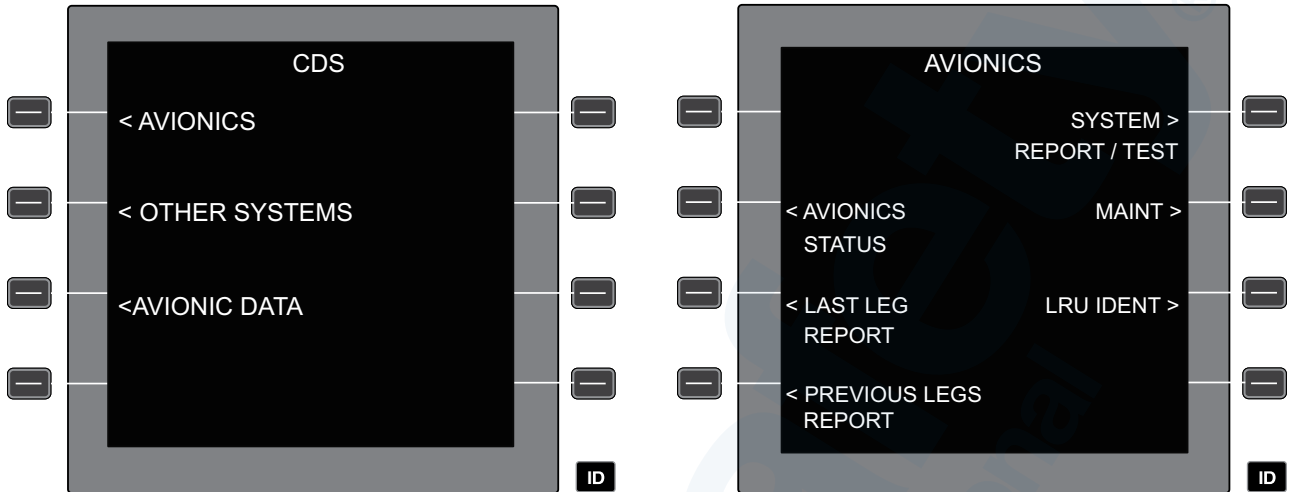
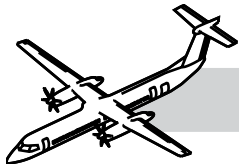


Figure 45-14. ARCDU, Avionics Page

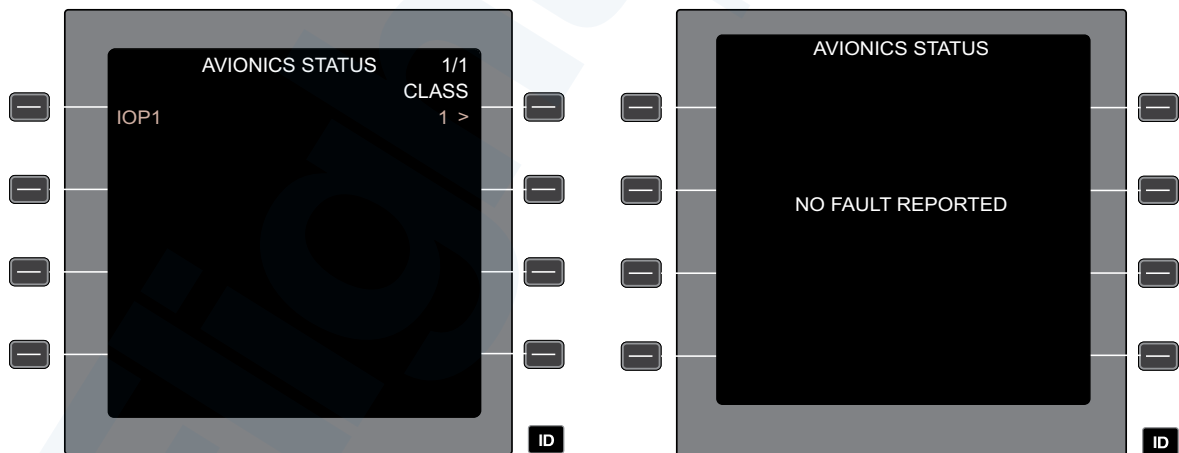
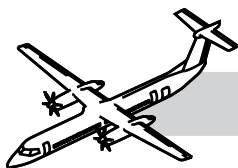


Figure 45-15. ARCDU, Avionics Status Page



A SYSTEM ERROR-ABORTED message is shown if incorrect data is received.

The CDS initial page has a menu for the selections of:

- AVIONICS
- OTHER SYSTEMS
- AVIONIC DATA.

Refer to Figure 45-14. ARCDU, Avionics Page.

The side key adjacent to AVIONICS on the CDS page is pushed to show the AVIONICS page.

The AVIONICS page has a menu for the selections of:

- AVIONICS STATUS
- LAST LEG REPORT
- PREVIOUS LEGS REPORT
- SYSTEM REPORT/TEST
- MAINT
- LRU IDENT.

NOTE

The non-volatile memory (NVM) of the CDS has the following in storage capacity:

- 50 class 1 failure messages in the LAST LEG REPORT
- 300 class 1 failure messages in the PREVIOUS LEGS REPORT
- 50 class 2 failure messages in MAINT REPORT.

Refer to Figure 45-15. ARCDU, Avionics Status Page.

The AVIONICS STATUS page shows current class 1 and class 2 internal malfunctions of avionic systems. To see current external faults, the side key adjacent to the cursor is pushed to go to the SYSTEM REPORT / TEST page. From this page, the side keys are pushed to start an IBIT to check for both internal and external malfunctions.

If a malfunction was sensed during the last leg, but is no longer present, the message is removed from the AVIONICS STATUS page.

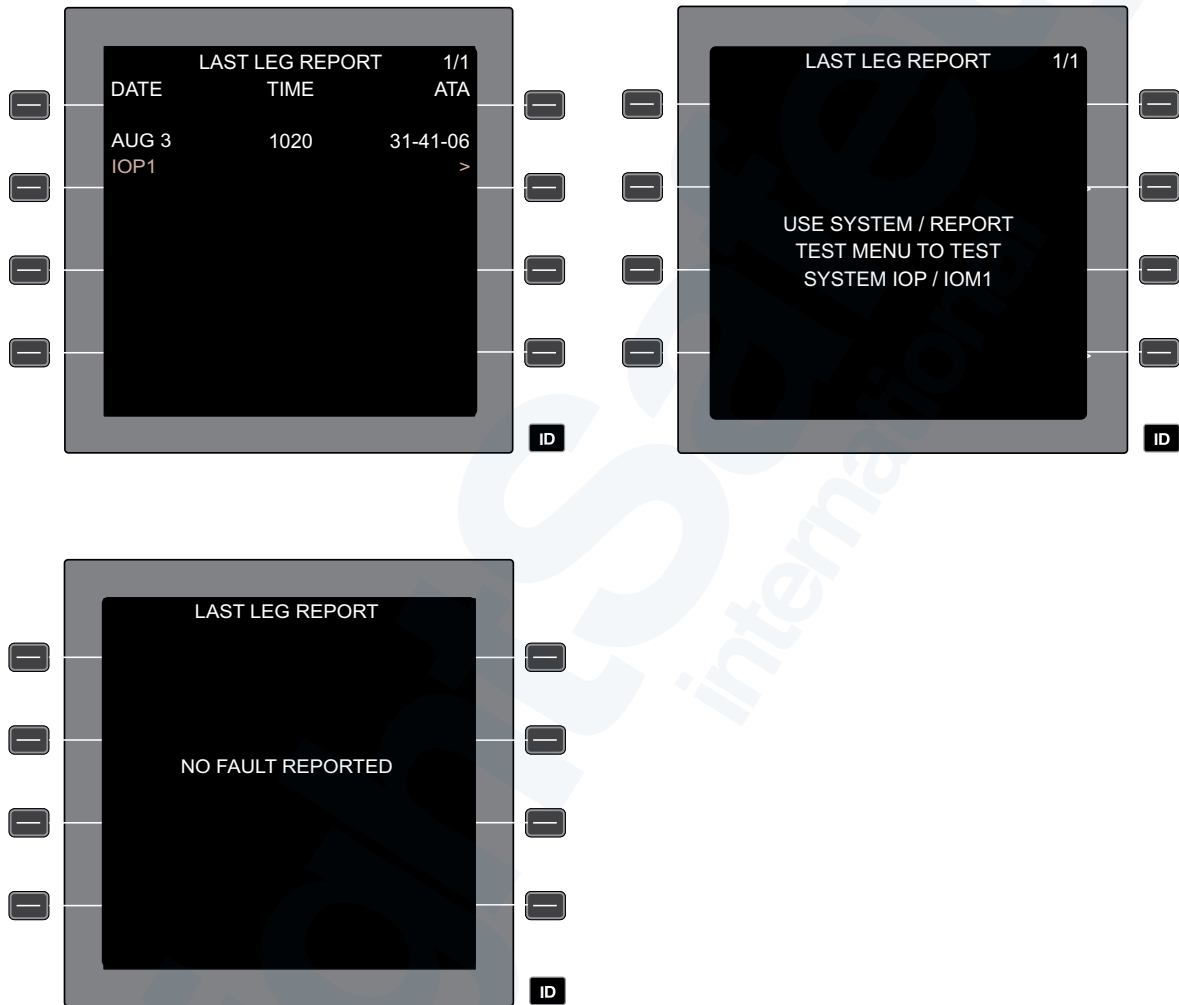
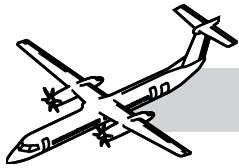
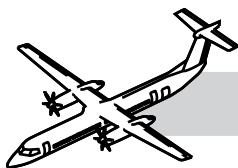


Figure 45-16. ARCDU, Last Leg Report Page



Refer to Figure 45-16. ARCDU, Last Leg Report Page.

NOTES

The LAST LEG REPORT page shows class 1 malfunctions sensed while the aircraft was airborne.

It shows the parameters that follow:

- Date
- Time
- ATA Message
- Up to 2 malfunctions can be shown on each LAST LEG REPORT page.

If the side key adjacent to the cursor is pushed, a message shown to USE SYSTEM REPORT/TEST MENU to start an IBIT for the related system.

At the start of a new leg when the CAS is more than 50 knots, all the messages are moved from the LAST LEG REPORT page to the PREVIOUS LEGS REPORT page.

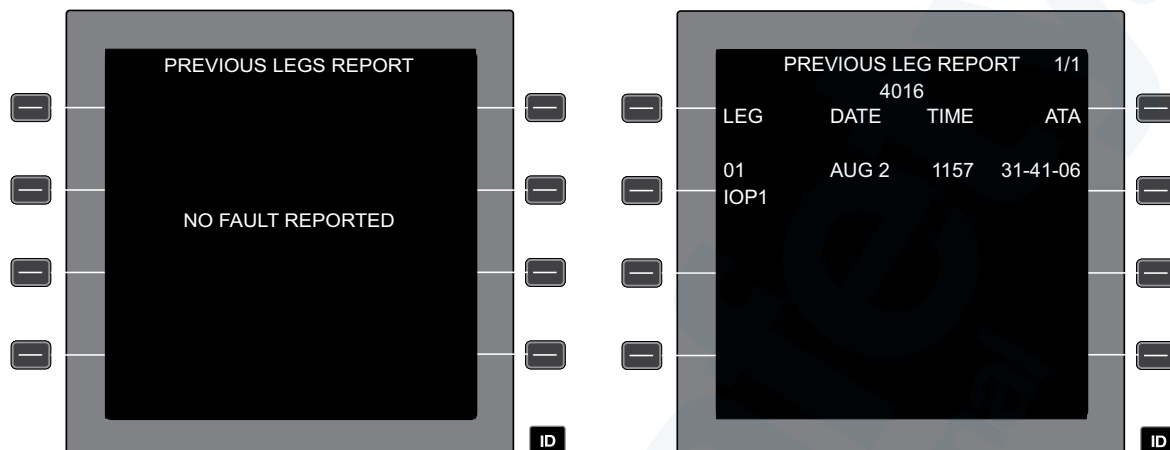
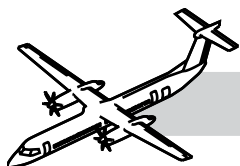


Figure 45-17. ARCDU, Previous Legs Report Page

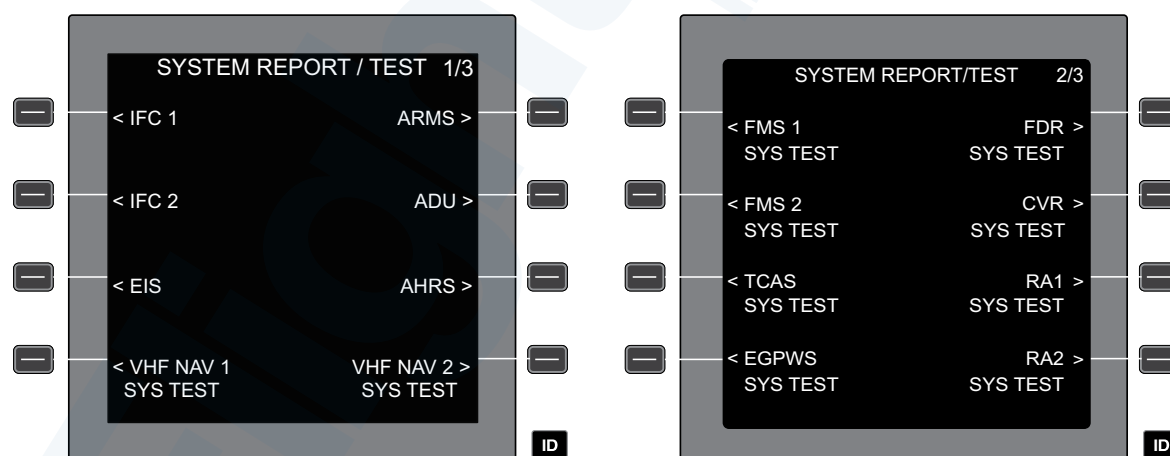
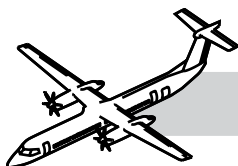


Figure 45-18. ARCDU, System Report/Test Page (Sheet 1 of 2)



Refer to Figure 45-17. ARCDU, Previous Legs Report Page.

The PREVIOUS LEGS REPORT page shows class 1 malfunctions sensed during the previous 63 legs. It shows the parameters that follow:

- Leg
- Date
- Time
- ATA
- Message.

Up to 2 malfunctions can be shown on each PREVIOUS LEGS REPORT page.

NOTE

Only malfunctions that occurred on the same aircraft (same aircraft serial number) are shown on the same page.

The number for the legs is shown in a reverse order so that the first shown failure is most recent.

The PREV and NEXT keys are used to see the different pages of the PREVIOUS LEGS REPORT.

Refer to Figure 45-18. ARCDU, System Report/Test Page (Sheet 1 of 2).

Page 1 of 3 of the SYSTEM REPORT / TEST pages has a menu for the selections of:

- IFC 1
- IFC 2
- EIS
- VHF NAV 1 SYS TEST
- ARMS
- ADU
- AHRS
- VHF NAV 2 SYS TEST.

Page 2 of 3 of the SYSTEM REPORT / TEST pages has a menu for the selections of:

- FMS 1 SYS TEST
- FMS 2 SYS TEST
- TCAS SYS TEST
- EGPWS SYS TEST
- FDR SYS TEST
- CVR SYS TEST
- RA1 SYS TEST
- RA2 SYS TEST.

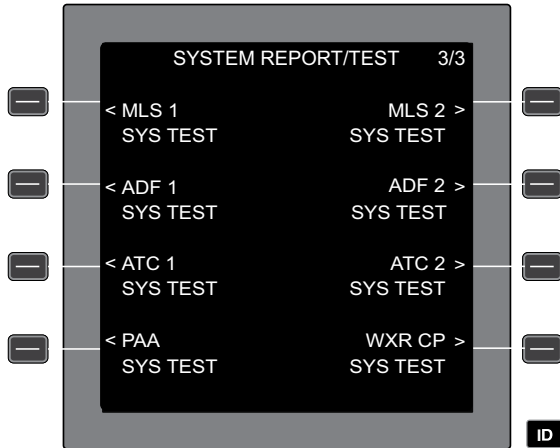
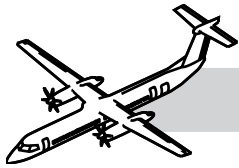


Figure 45-19. ARCDU, System Report/Test Page (Sheet 2 of 2)

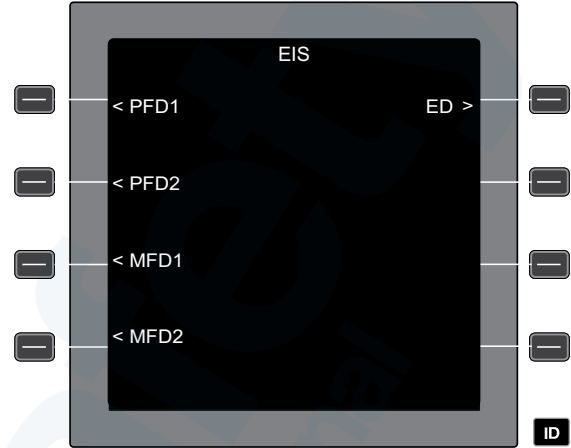


Figure 45-21. ARCDU, System Report/Test, EIS Page

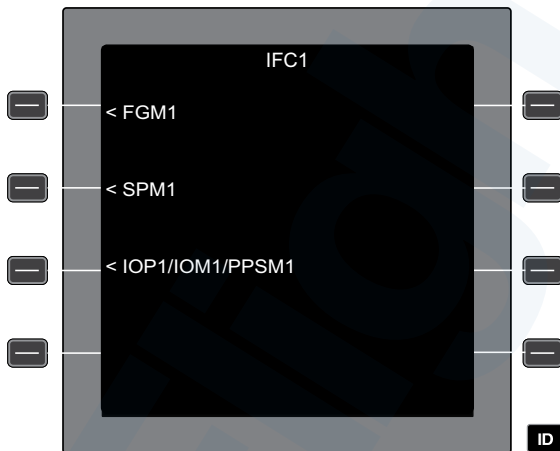


Figure 45-20. ARCDU, System Report/Test, IFC Page

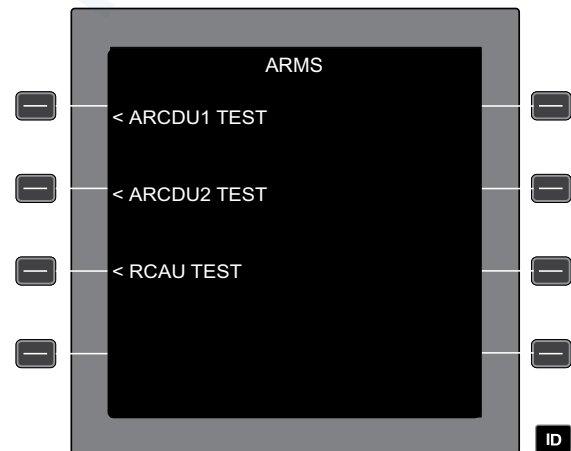
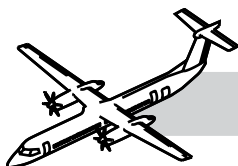


Figure 45-22. ARCDU, System Report/Test, Arms Page



Refer to Figure 45-19. ARCDU, System Report/Test Page (Sheet 2 of 2).

Page 3 of 3 of the SYSTEM REPORT / TEST pages has a menu for the selections of:

- MLS 1 SYS TEST
- ADF 1 SYS TEST
- ATC 1 SYS TEST
- PAA SYS TEST
- MLS 2 SYS TEST
- ADF 2 SYS TEST
- ATC 2 SYS TEST
- WXR CP SYS TEST.

Refer to Figure 45-20. ARCDU, System Report/Test, IFC Page.

The side key adjacent to IFC on the SYSTEM REPORT / TEST page is pushed to show the IFC menu selections:

- FGM 1
- SPM 1
- IOP 1/IOM 1/PPSM 1.

Refer to Figure 45-21. ARCDU, System Report/Test, EIS Page.

The side key adjacent to the EIS on the SYSTEM REPORT / TEST page is pushed to show the EIS menu selections:

- PFD 1
- PFD 2
- MFD 1
- MFD 2
- ED.

Refer to Figure 45-22. ARCDU, System Report/Test, Arms Page

The side key adjacent to the ARMS on the SYSTEM REPORT / TEST page is pushed to show the ARMS menu selections:

- ARCDU 1 TEST
- ARCDU 2 TEST
- RCAU TEST.

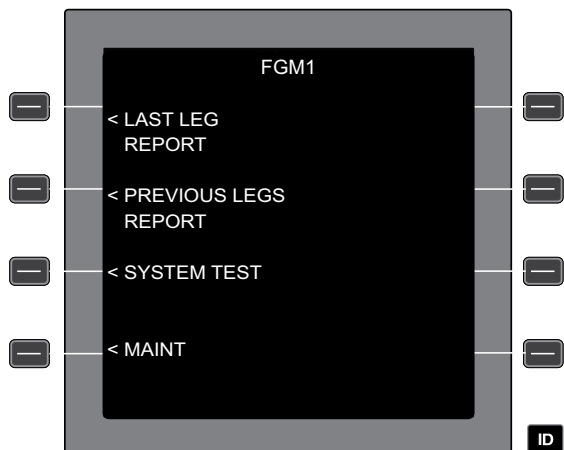
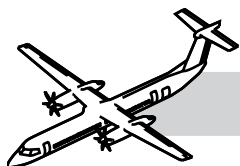


Figure 45-23. ARCDU, System Report/Test, FGM 1 Page



Figure 45-25. ARCDU, System Report/Test, FGM 1 Previous Legs Report Page

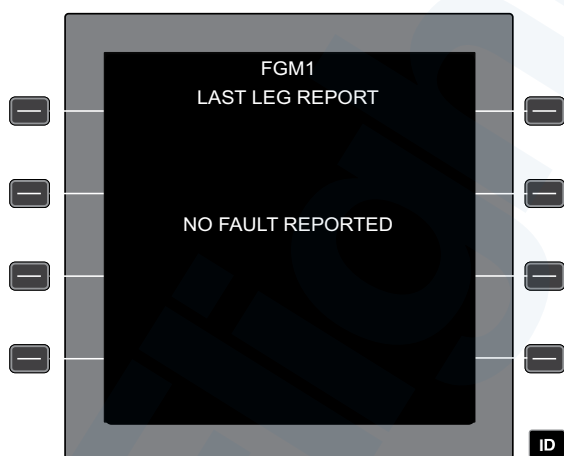
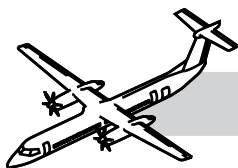


Figure 45-24. ARCDU, System Report/Test, FGM 1 Last Leg Report Page



Refer to Figure 45-23. ARCDU, System Report/Test, FGM 1 Page.

NOTES

The side key adjacent to FGM 1 on the SYSTEM REPORT/TEST page is pushed to show the FGM 1 menu selections:

- LAST LEG REPORT
- PREVIOUS LEGS REPORT
- SYSTEM TEST
- MAINT.

NOTE

These selections are available for all systems that have a type 1 interface with the CDS.

Refer to Figure 45-24. ARCDU, System Report/Test, FGM 1 Last Leg Report Page.

The side key adjacent to LAST LEG REPORT on the FGM 1 page is pushed to show the FGM 1 LAST LEG REPORT page.

Refer to Figure 45-25. ARCDU, System Report/Test, FGM 1 Previous Legs Report Page.

The side key adjacent to PREVIOUS LEGS REPORT on the FGM 1 page is pushed to show the FGM 1 PREVIOUS LEGS REPORT page.

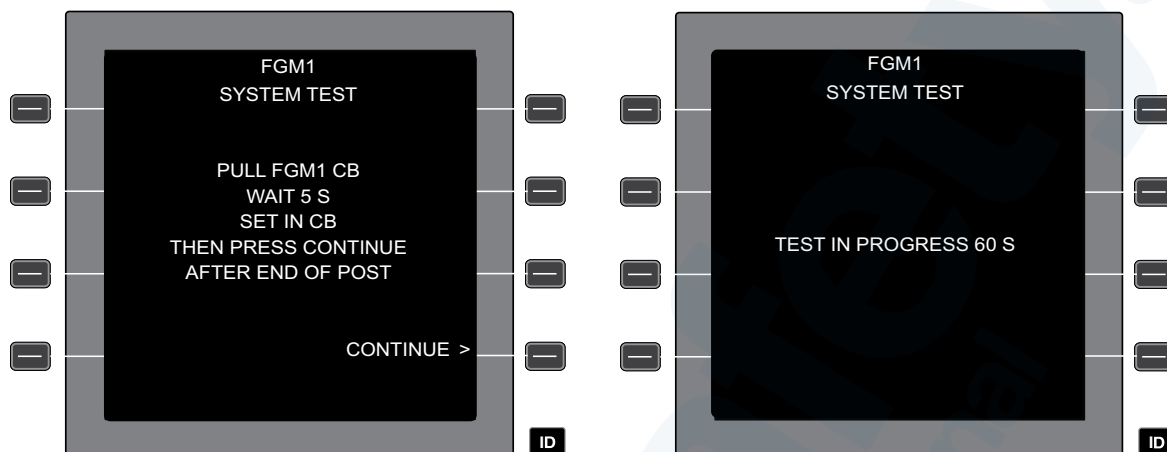
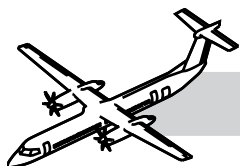


Figure 45-26. ARCDU, System Report/Test, FGM 1 System Test Page, Sheet 1

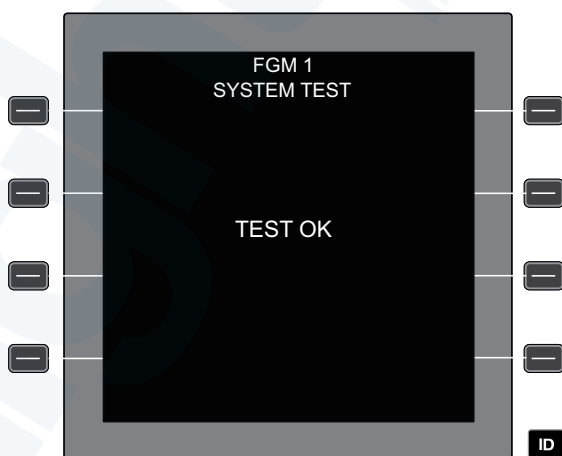
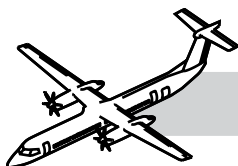


Figure 45-27. ARCDU, System Report/Test, FGM 1 System Test Page, Sheet 2



Refer to Figure 45-26. ARCDU, System Report/Test, FGM 1 System Test Page, Sheet 1.

NOTES

The side key adjacent to SYSTEM TEST on the FGM 1 page is pushed to show the FGM 1 SYSTEM TEST page.

When the conditions to start the IBIT are set, the side key adjacent to CONTINUE is pushed to continue the sequence.

A TEST IN PROGRESS 60S message is shown during the test.

Refer to Figure 45-27. ARCDU, System Report/Test, FGM 1 System Test Page, Sheet 2.

If the IBIT is not satisfactory, the SYSTEM TEST page shows the following parameters:

- ATA of the line replaceable unit (LRU)
- Class of malfunction, 1 or 2
- Message.

If the test is satisfactory, TESK OK is shown.

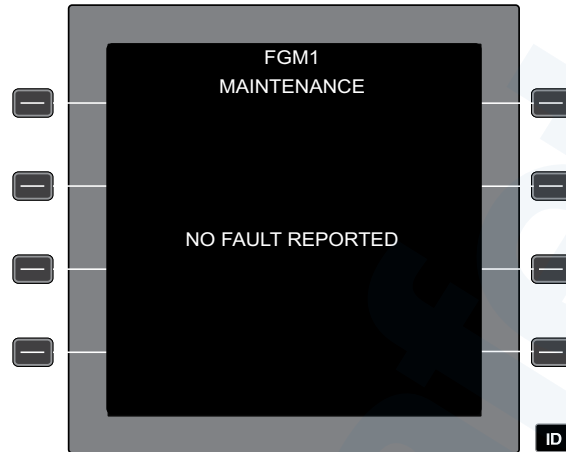
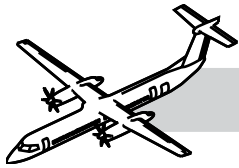


Figure 45-28. ARCDU, System Report/Test, FGM 1 Maint Page

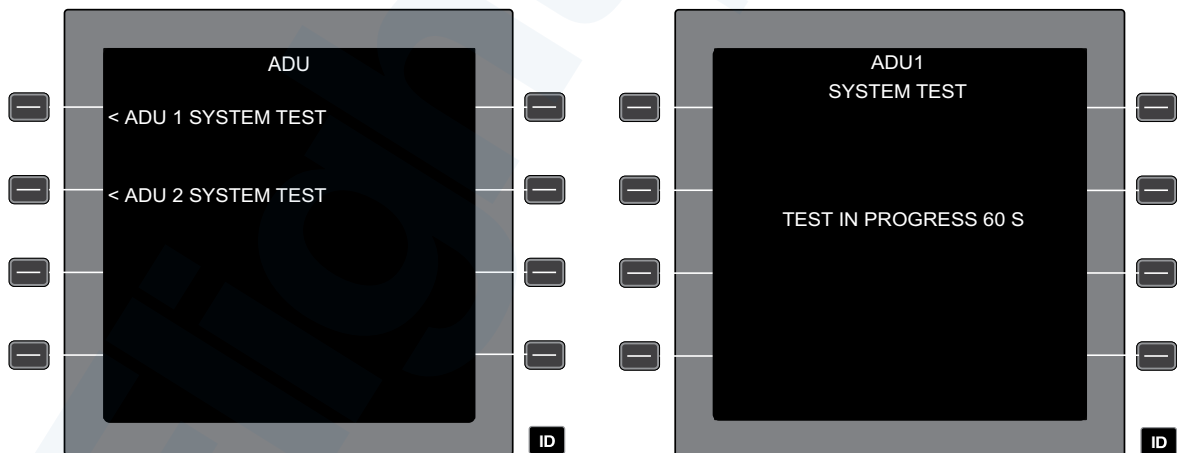
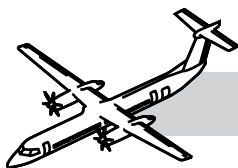


Figure 45-29. ARCDU, System Report/Test, ADU Page



Refer to Figure 45-28. ARCDU, System Report/Test, FGM 1 Maint Page.

NOTES

The side key adjacent to MAINT on the FGM 1 page is pushed to show the FGM 1 MAINT page.

Refer to Figure 45-29. ARCDU, System Report/Test, ADU Page.

The side key adjacent to ADU 1 SYSTEM TEST on the ADU page is pushed to start the IBIT.

The ADU 1 SYSTEM TEST page shows a TEST IN PROGRESS 60S message during the test.

NOTE

These selections are similar for all systems that have a type 2 interface with the CDS.

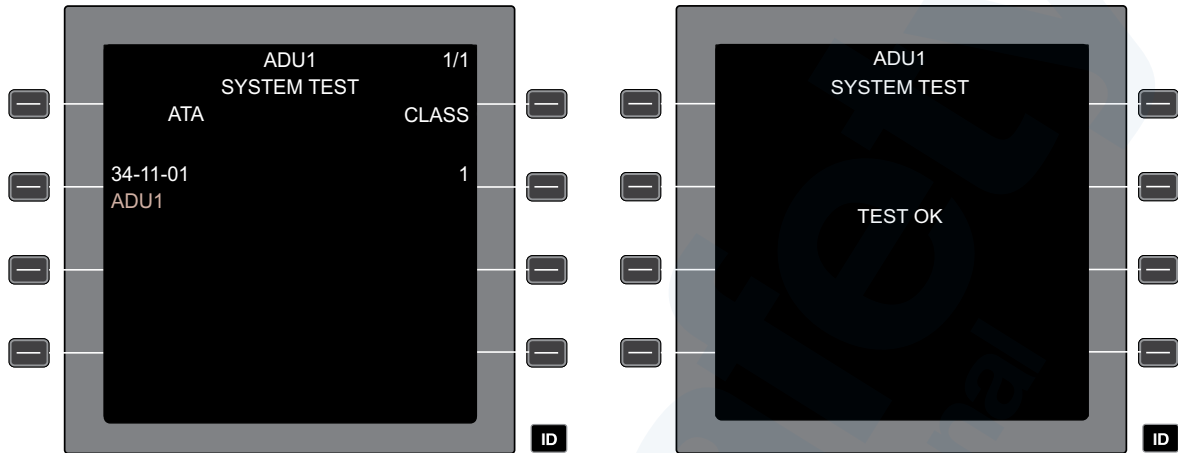
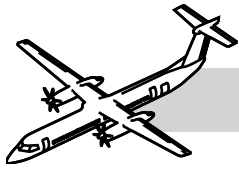


Figure 45-30. ARCDU, System Report/Test, ADU 1 System Test Page

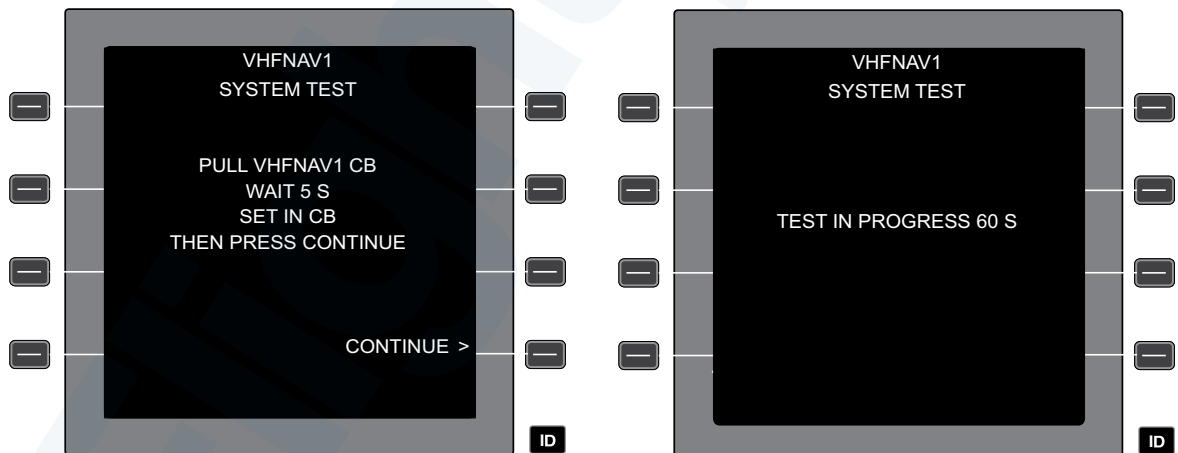
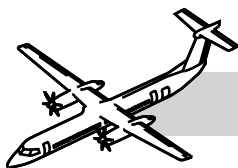


Figure 45-31. ARCDU, System Report/Test, VHF NAV 1 System Test Page



Refer to Figure 45-30. ARCDU, System Report/Test, ADU 1 System Test Page.

NOTES

If the IBIT is not satisfactory, the SYSTEM TEST page shows the following parameters:

- ATA of the line replaceable unit (LRU)
- Class of malfunction, 1 or 2
- Message.

If the test is satisfactory, TEST OK is shown.

Refer to Figure 45-31. ARCDU, System Report/Test, VHF NAV 1 System Test Page.

The side key adjacent to VHF NAV 1 SYSTEM TEST on the SYSTEM REPORT / TEST page is pushed to show the VHF NAV 1 SYSTEM TEST page.

The circuit breaker for the system is pulled and then set in. The side key adjacent to CONTINUE is then pushed to continue the sequence.

The VHF NAV 1 SYSTEM TEST page shows A TEST IN PROGRESS 60S message during the test.

NOTE

These selections are similar for all systems that have a type 3 interface with the CDS.

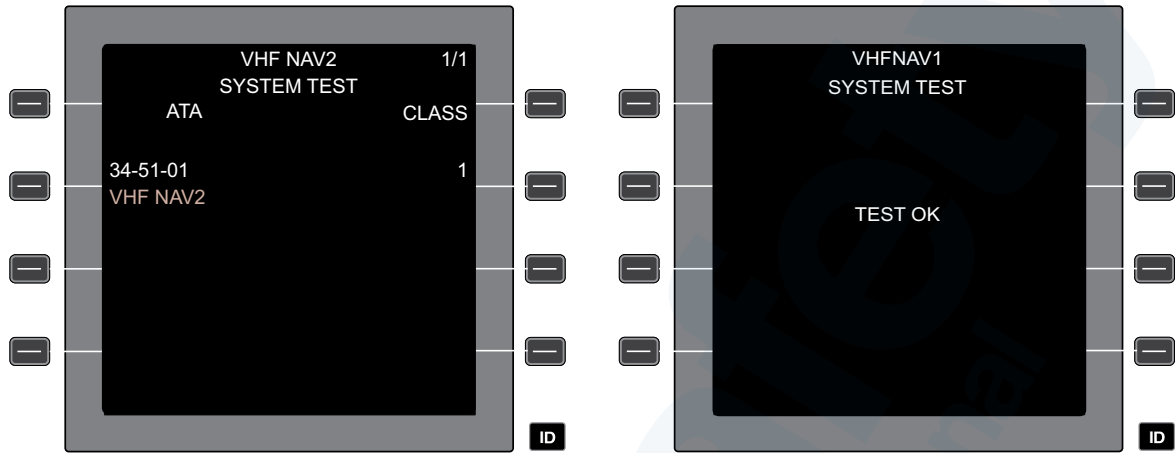
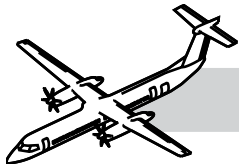


Figure 45-32. ARCDU, System Report/Test, VHF NAV 1 System Test Page

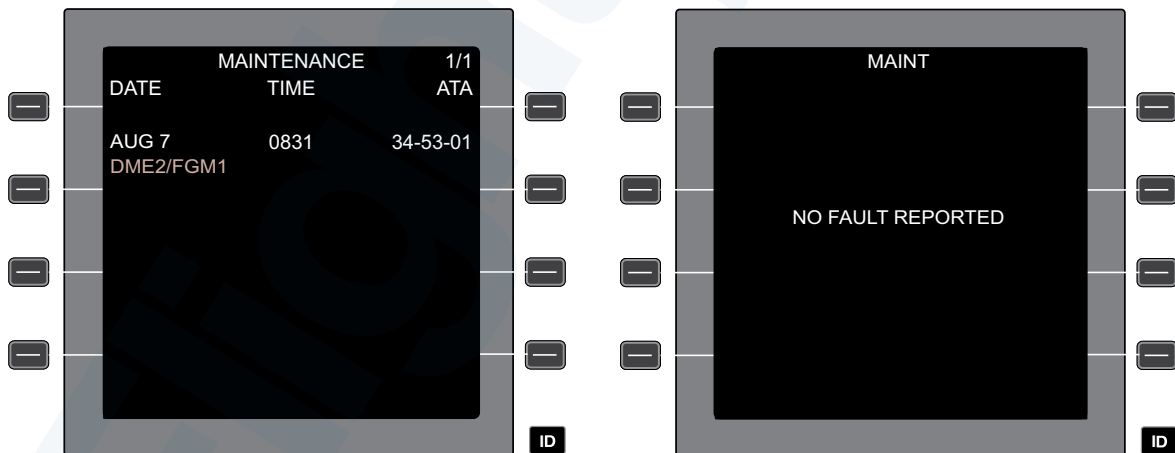
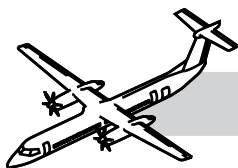


Figure 45-33. ARCDU, Maintenance Page



Refer to Figure 45-32. ARCDU, System Report/Test, VHF NAV 1 System Test Page.

NOTES

If the system test is not satisfactory, the SYSTEM TEST page shows the following parameters:

- ATA of the line replaceable unit (LRU)
- Class of malfunction, 1 or 2
- Message.

If the system is satisfactory, TEST OK is shown.

Refer to Figure 45-33. ARCDU, Maintenance Page.

The MAINTENANCE page shows class 2 malfunctions sensed while the aircraft was airborne.

It shows the following parameters:

- Date
- Time
- ATA
- Message.

Up to 2 malfunctions can be shown on the same MAINTENANCE page.

At the start of a new leg when the CAS is more than 50 knots, all class 2 messages are erased.

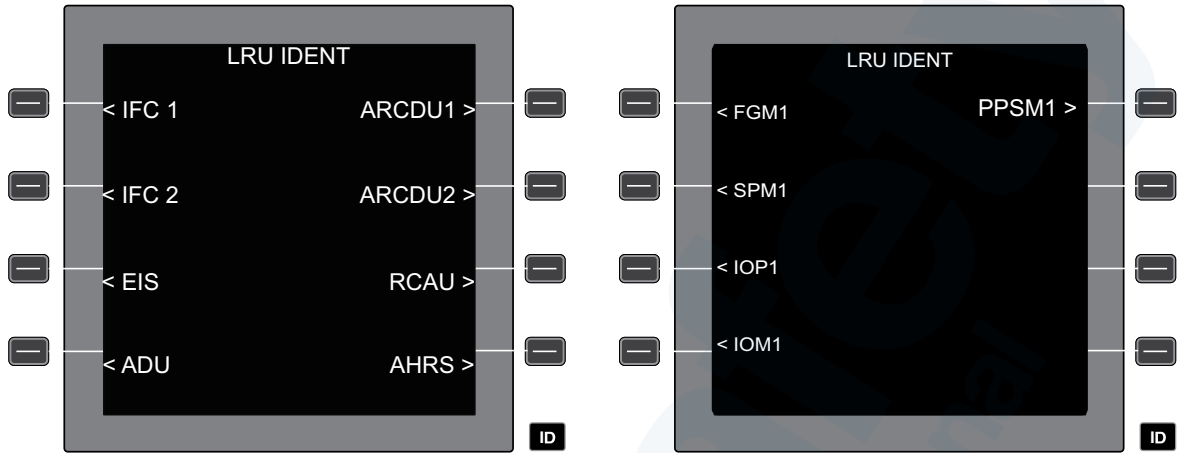
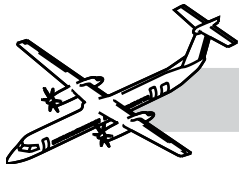


Figure 45-34. ARCDU, LRU Ident Pages Sheet 1

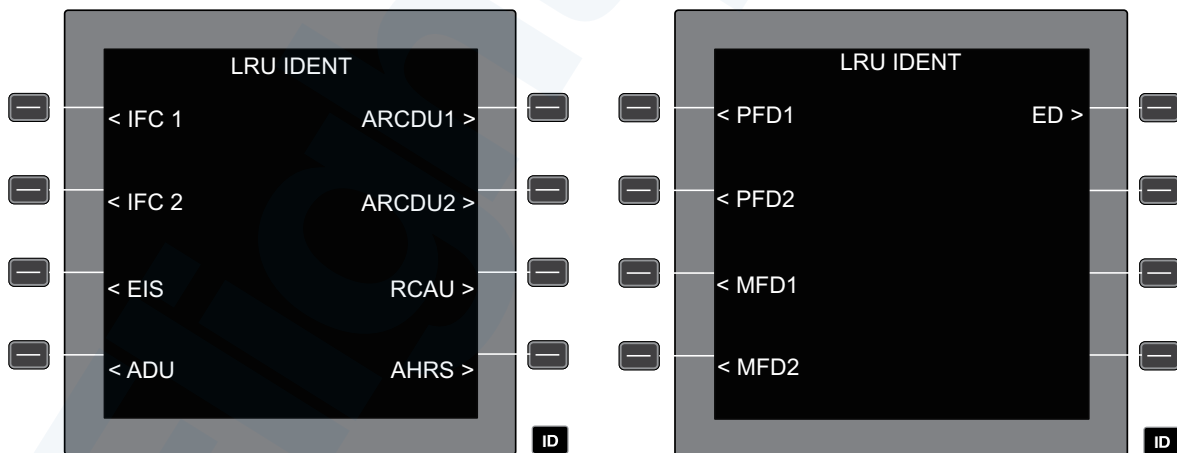
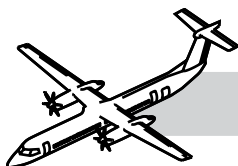


Figure 45-35. ARCDU, LRU Ident Pages Sheet 2



Refer to Figure 45-34. ARCDU, LRU Ident Pages Sheet 1.

The first LRU IDENT page has a menu for selections of:

- IFC 1
- IFC 2
- EIS
- ADU
- ARCDU 1
- ARCDU 2
- RCAU
- AHRS.

For the IFC 1, the LRU IDENT page has a menu for selections of:

- FGM 1
- SPM 1
- IOP 1
- IOM 1
- PPSM 1.

For the IFC 2, the LRU IDENT page has a menu for selections of:

- FGM 2
- SPM 2
- IOP 2
- IOM 2
- PPSM 2.

Refer to Figure 45-35. ARCDU, LRU Ident Pages Sheet 2.

For the EIS, the LRU IDENT page has a menu for the selections of:

- PFD 1
- PFD 2
- MFD 1
- MFD 2
- ED

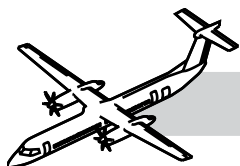


Figure 45-36. ARCDU, LRU Ident Pages Sheet 3

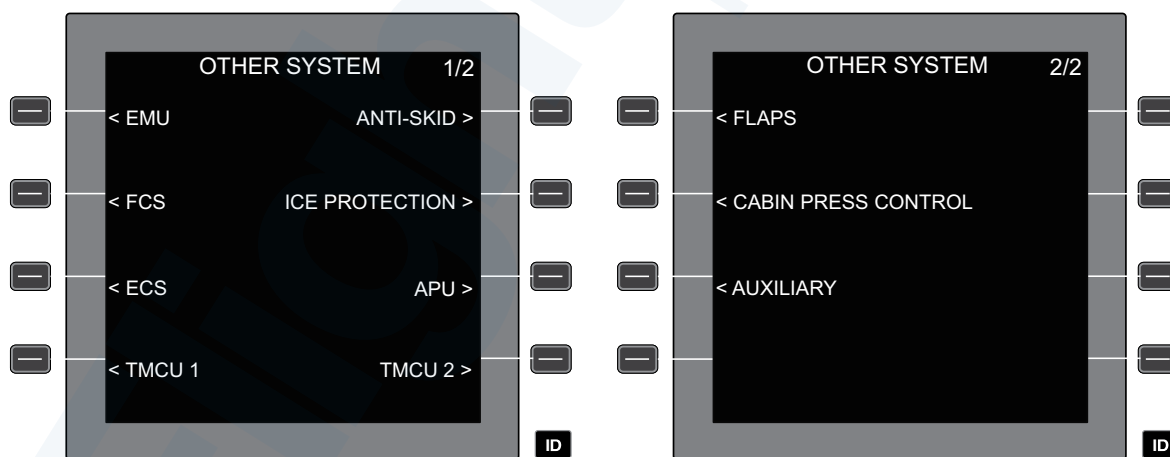
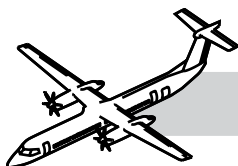


Figure 45-37. ARCDU, Other System Page



Refer to Figure 45-36. ARCDU, LRU Ident
Pages Sheet 3.

NOTES

The LRU IDENT page shows the part number
page for the LRU.

Refer to Figure 45-37. ARCDU, Other System
Page.

Page 1 of 2 of the OTHER SYSTEM pages has
a menu for selections of:

- EMU
- FCS
- ECS
- TMCU 1
- TMCU 2
- ANTI-SKID
- ICE PROTECTION
- APU.

Page 2 of 2 of the SYSTEM REPORT / TEST
pages has a menu for the selections of:

- FLAPS
- CABIN PRESSURE CONTROL
- AUXILIARY.

For fault retrieval and diagnostic detail of the
above systems, please refer to their relevant
ATA chapters in the Maintenance Training
Manual.

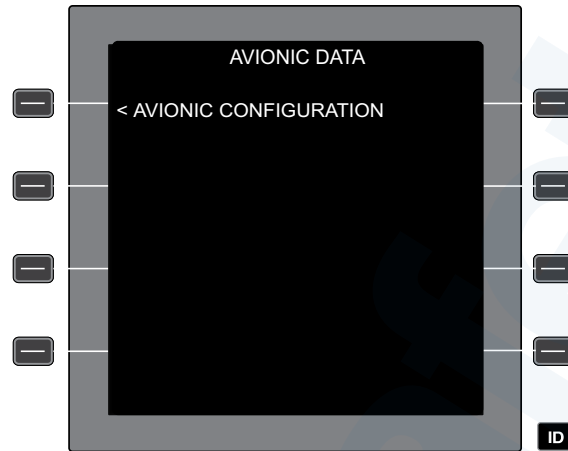
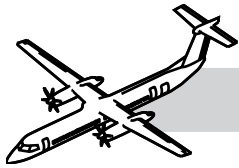
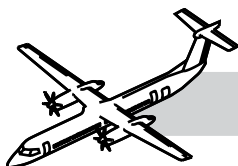


Figure 45-38. ARCDU, Avionics Data Page

AVIONIC CONFIG 1/8			AVIONIC CONFIG 2/8		
	ACM1	ACM2		ACM1	ACM2
TCAS	YES	YES	HF 1	YES	YES
ACARS	YES	YES	HF 2	NO	NO
SELCAL	YES	YES	VHF 1 EXT RGE	NO	NO
MLS	YES	YES	VHF 2 EXT RGE	NO	NO
FMS1	YES	YES	VHF 3 EXT RGE	NO	NO
FMS2	YES	YES	VHF 8.33 KHZ	YES	YES
VHF COM3	NO	NO	DME2	YES	YES
TACAN	NO	NO	ATC2	YES	YES
UHF	NO	NO	ADF2	YES	YES
VHF FM	NO	NO			

Figure 45-39. ARCDU, Avionics Config Sheet 1



Refer to Figure 45-38. ARCDU, Avionics Data Page.

NOTES

The AVIONICS DATA page has a menu to show the AVIONICS CONFIGURATION pages.

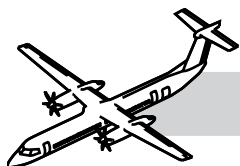
Refer to:

- Figure 45-39. ARCDU, Avionics Config Sheet 1.
- Figure 45-40. ARCDU, Avionics Config Sheet 2.

Eight pages of the AVIONICS CONFIG pages show YES or NO for the programming of the aircraft configuration modules (ACM1 and ACM2). It also shows the cyclic redundancy checks (CRC1 and CRC2).

NOTE

A cyclic redundancy check (CRC) is a non-secure hash function designed to detect accidental changes to raw computer data. A CRC-enabled device calculates a fixed-length binary sequence, known as the CRC code for each block of data and stores it. When a block is received the device repeats the calculation; if the new CRC does not match the one calculated earlier, then the block contains a data error and the device takes corrective action such as rereading or requesting the block be sent again.

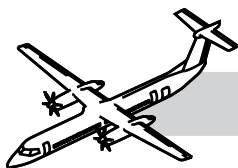


AVIONIC CONFIG 3/8			AVIONIC CONFIG 4/8		
	ACM1	ACM2		ACM1	ACM2
APU	YES	YES	AIRCRAFT TYPE :	400	400
HUD	NO	NO	DASH 8 -		
AUX NAV 1	NO	NO	ENGINE TYPE :	PWC	PWC
AUX NAV 2	NO	NO	UNITS :	KG	KG
RA2	YES	YES		L	L
IRS	NO	NO		M	M
EGPWS	YES	YES		C	C
ALTERNATE ICP	NO	NO	A/C NUMBER :	4001	4001
COUPLED HEADING	NO	NO			
FLT TEST A/C	YES	YES			

AVIONIC CONFIG 5/8			AVIONIC CONFIG 6/8		
	ACM1	ACM2		ACM1	ACM2
AOA 1 CORRECTION :		+0.000	DISPLAY FORMAT :	1CUE	1CUE
AOA 2 CORRECTION :		+0.000			

AVIONIC CONFIG 7/8			AVIONIC CONFIG		
	ACM1	ACM2		ACM1	ACM2
GPWS	YES	YES	TONES:		
TCAS	YES	YES	MAST WARN	YES	YES
FIRE BELL	YES	YES	MAST CAUTION	YES	YES
TO CONFIG	YES	YES			
AP DIENG	YES	YES			
PITCH TRIM	YES	YES			
OVERSPEED	YES	YES	CRC1 : XXXXXXXX		
INCORRECT LD	YES	YES			
ALTITUDE ALRT	YES	YES	CRC2 : XXXXXXXX		
BETA LOCKOUT	YES	YES			
SELCAL	YES	YES			

Figure 45-40. ARCDU, Avionics Config Sheet 2



Refer to Figure 45-41. Maintenance Panel, NVM Reset of Other Systems.

It is usually necessary to clear the CDS messages after they are seen on the ARCDU.

The messages for the avionic systems are automatically reset when the defective system is repaired.

A manual reset is necessary for other systems.

To reset the NVM of the other systems, the knob for the other systems is set from OFF to a position that follows:

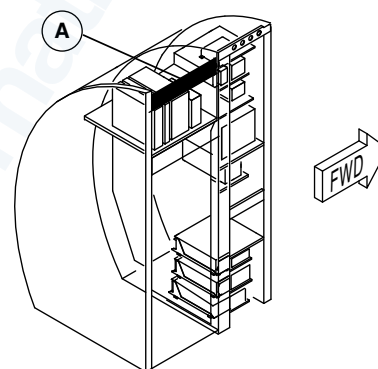
- FLAP
- PROP DEICE 2
- PROP DEICE 1
- AFR DEICE
- FLT CTL
- ECS
- CPC.

The amber ON light adjacent to the knob illuminates.

Then, the NVM switch is pushed.

NOTE

The CDS does not need to be set to maintenance mode to reset messages for other systems. But, the CDS maintenance mode should be set to see that the message is reset.



AVIONICS RACK ELECTRICAL
INSTALLATION

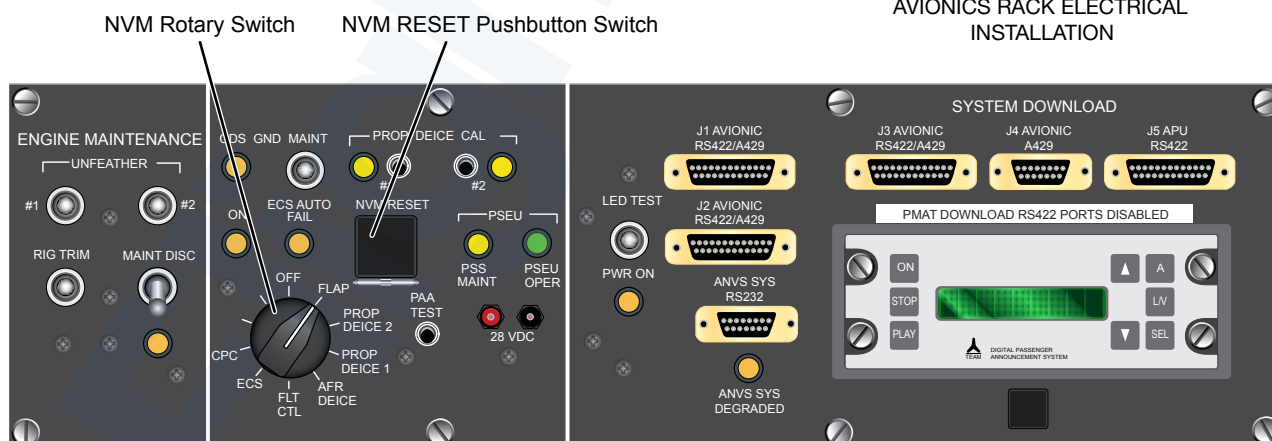


Figure 45-41. Maintenance Panel, NVM Reset of Other Systems

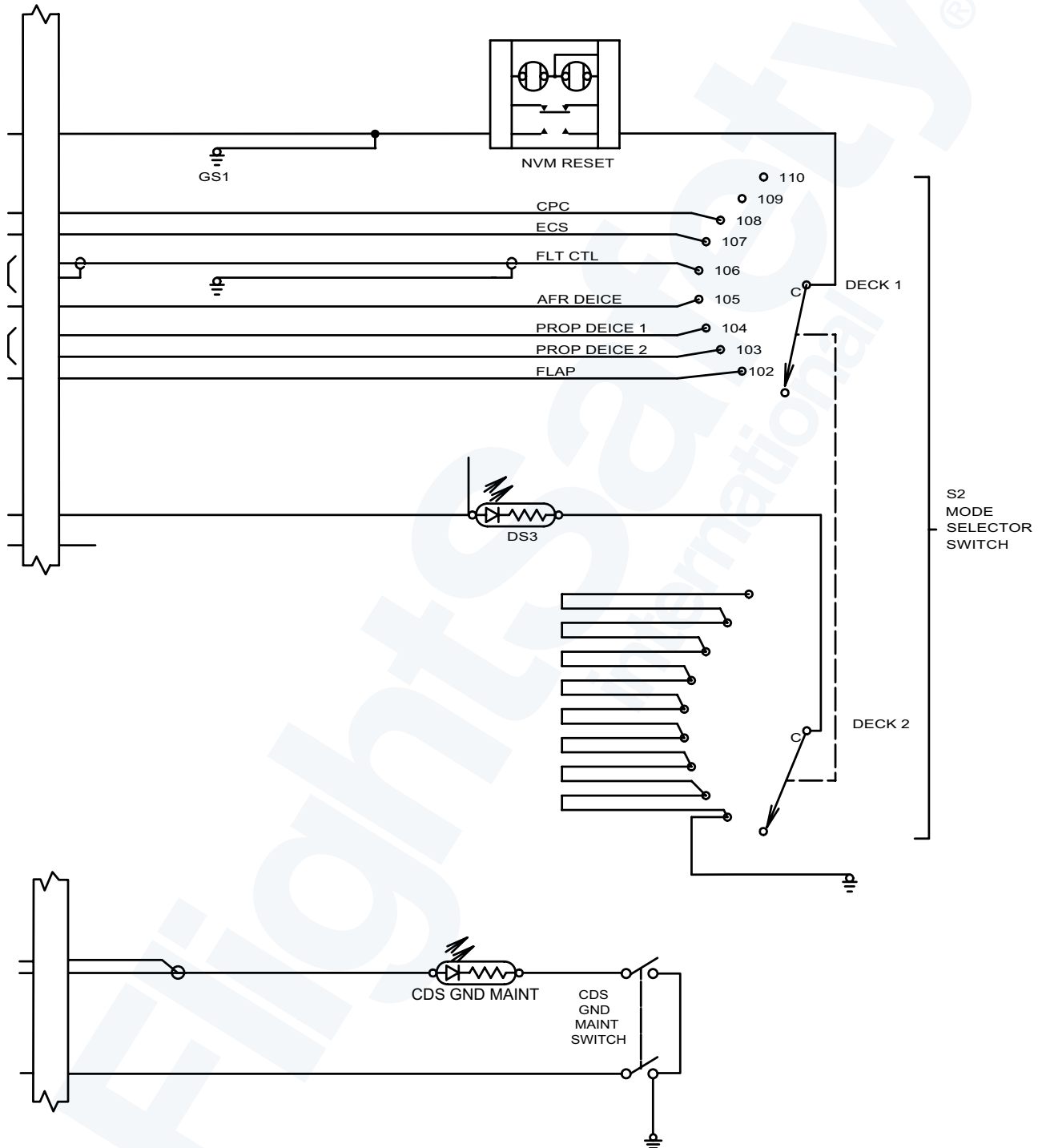
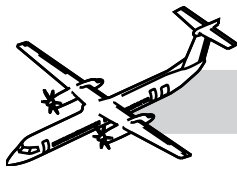
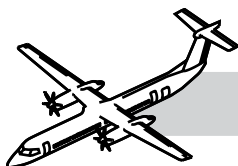


Figure 45-42. CDS Operation, Maintenance Control Panel



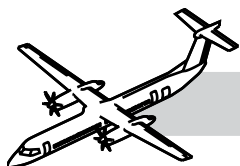
OPERATION

NOTES

Refer to Figure 45-42. CDS Operation, Maintenance Control Panel.

A ground signal from the CDS GND MAINT switch is supplied to the two ARCDUs and the IOP 1 to start the maintenance mode of the CDS and make the ON light on the maintenance control panel illuminate.

A ground signal is supplied through the NVM RESET switch and mode selector switch to reset the NVM of the unit.



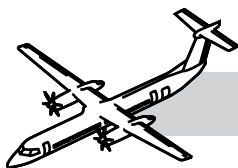
45-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- DB9F to DB25M Cable, Download
- DB25F Connector/Backshell, EMU-clear Hardware Interlock
- GBS Download Computer

45-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- FIM 45-45-00-810-802 Centralized Diagnostics System (CDS) Fault Message Retrieval – Fault Isolation
- AMM 45-45-00-710-801 Operational Test for the Return-To-Service of the Maintenance Panel
- AMM 45-00-21-742-801 Operational Test for the Return-To-Service of the Maintenance Panel
- AMM 45-00-21-743-801 Erase the Data from the Central Diagnostic System (CDS) – Environmental Control System (ECS)
- AMM 45-00-21-742-802 Retrieval of Data from the Central Diagnostic System (CDS) – Cabin Pressure Control System (CABIN PRESS CONTROL)
- AMM 45-00-21-743-802 Erase the Data from the Central Diagnostic System (CDS) – Cabin Pressure Control System (CABIN PRESS CONTROL)
- AMM 45-00-27-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Flight Control System (FCS)
- AMM 45-00-27-743-801 Erase the Data from the Central Diagnostic System (CDS) – Flight Control System (FCS)
- AMM 45-00-27-742-802 Retrieval of Data from the Central Diagnostic System (CDS) – Flaps System (FLAPS)
- AMM 45-00-27-743-802 Erase the Data from the Central Diagnostic System (CDS) – Flaps System (FLAPS)
- AMM 45-00-30-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Timer Monitor Control Unit (TMCU)
- AMM 45-00-30-743-801 Erase the Data from the Central Diagnostic System (CDS) – Timer Monitor Control Unit (TMCU)
- AMM 45-00-30-742-802 Retrieval of Data from the Central Diagnostic System (CDS) – Ice Protection System (ICE PROTECTION)
- AMM 45-00-30-743-802 Erase the Data from the Central Diagnostic System (CDS) – Ice Protection System (ICE PROTECTION)
- AMM 45-00-32-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Anti-Skid System (ANTI-SKID)
- AMM 45-00-32-743-801 Erase the Data from the Central Diagnostic System (CDS) – Anti-Skid System (ANTI-SKID)



- AMM 45-00-34-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Avionics (AVIONICS)
- AMM 45-00-34-742-802 Retrieval of Data from the Central Diagnostic System (CDS) – Avionic Data (AVIONIC DATA)
- AMM 45-00-49-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Auxiliary Power Unit (APU)
- AMM 45-00-49-743-801 Erase the Data from the Central Diagnostic System (CDS) – Auxiliary Power Unit (APU)
- AMM 45-00-73-742-801 Retrieval of Data from the Central Diagnostic System (CDS) – Engine Monitoring (EMU)
- AMM 45-00-73-743-801 Erase the Data from the Central Diagnostic System (CDS) – Engine Monitoring Unit (EMU)